MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED SCHOTTKY TTL, OCTAL BUFFERS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, advanced Schottky TTL, octal buffer (3-state) microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device types. The device types shall be as follows:

Device type	Circuit
01	Octal buffer with active low enabled 3-state inverted outputs
02	Octal buffer with active low and active high enabled 3-state noninverted outputs
03	Octal buffer with active low enabled 3-state noninverted outputs
04	Octal buffer with active low enabled 3-state inverted outputs (inputs and outputs opposite side of package)

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
R	D-8 (20-lead, 1/4" x 1 1/16"), dual-in-line package
S	F-9 (20-lead, 1/4" x 1/2"), flat package
X	C-2A (20-terminal, .350" x .350") square chip carrier package
2	C-2 (20-terminal, .350" x .350") square chip carrier package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V to +7.0 V
Input voltage range	-1.2 V at -18 mA to +7.0 V
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) 1/:	
Device types 01 and 04	412 mW
Device types 02 and 03	495 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Cases R, S, X, and 2	(See MIL-M-38510, appendix C)
Junction temperature (T_J) 2/	+175°C

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).
2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions in accordance with method 5004 of MIL-STD-883.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V minimum to 5.5 V maximum
Minimum high level input voltage (V_{IH})	2.0 V
Maximum low level input voltage (V_{IL})	0.8 V
Normalized fanout (each output): 3/	
At low logic level	33 maximum
At high logic level	50 maximum
Case operating temperature range (T_C)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specification and standard, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Logic diagrams and truth tables. The logic diagrams and truth tables shall be as specified on figure 2.

3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers schematics shall be maintained by the agent activity and will be available upon request.

3/ The device shall fanout in both high and low levels to the specified number of inputs of the same device type as that being tested.

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3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.4).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$ unless otherwise specified	Device type	Limits		Unit
				Min	Max	
High level output voltage	V_{OH}	V_{CC} = 4.5 V, I_{OH} = -3.0 mA, V_{IL} = 0.8 V, V_{IH} = 2.0 V	All	2.4		V
Low level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 48 mA, V_{IL} = 0.8 V, V_{IH} = 2.0 V	All		0.55	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA, T_C = 25°C	All		-1.2	V
High level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	All		20	µA
	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 7.0 V	All		100	µA
Low level input current	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.5 V	01	-.03	-1.0	mA
			04	-.25	-.60	mA
Low level input current (at data)	I_{IL1}		02,03	-.03	-1.6	mA
Low level input enable current	I_{IL2}		02,03	-.03	-1.0	mA
Short-circuit output 1/ current	I_{OS}	V_{CC} = 5.5 V	All	-100	-375	mA
Output drive	I_{OD}	V_{CC} = 4.5 V, V_{IN} = 5.5 V, V_{OUT} = 2.5 V	01,02, 03	65		mA
High level supply current	I_{CCH}	V_{CC} = 5.5 V	01		35	mA
			02,03		60	mA
			04		20	mA
Low level supply current	I_{CCL}	V_{CC} = 5.5 V	01,04		75	mA
			02,03		90	mA
Off state supply current	I_{CCZ}	V_{CC} = 5.5 V; outputs disabled	01		75	mA
			02,03		90	mA
			04		45	mA

See footnote at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$ unless otherwise specified	Device type	Limits		Unit
				Min	Max	
Off state output leakage current	I_{OZH}	V_{CC} = 5.5 V, V_{ZH} = 2.4 V	All		50	µA
	I_{OZL}	V_{CC} = 5.5 V, V_{ZL} = 0.5 V	All		-50	mA
Propagation delay time, low-to-high level, A to $\overline{Y}$	t_{PLH1}	V_{CC} = 5.0 V, C_L = 50 pF ±10%, R_L = 500Ω ±5% (see figure 4)	01	1.0	9.0	ns
	t_{PLH3}		04	1.0	6.0	ns
Propagation delay time, low-to-high level, A to Y	t_{PLH2}		02,03	1.0	6.5	ns
Propagation delay time, high-to-low level, A to $\overline{Y}$	t_{PHL1}		01	1.0	6.0	ns
	t_{PHL3}		04	1.0	4.5	ns
Propagation delay time, high-to-low level, A to Y	t_{PHL2}		02,03	1.0	7.0	ns
Propagation delay time, low level to off state, $\overline{OE}$ to $\overline{Y}$	t_{PLZ1}		01	2.0	12.5	ns
	t_{PLZ4}		04	1.0	7.5	ns
Propagation delay time, low level to off state, $\overline{OE}$ to Y	t_{PLZ2}		02,03	2.0	7.5	ns
Propagation delay time, low level to off state, OE to Y	t_{PLZ3}		02	2.0	7.5	ns
Propagation delay time, high level to off state, $\overline{OE}$ to $\overline{Y}$	t_{PHZ1}		01	2.0	6.5	ns
	t_{PHZ4}		04	1.5	7.0	ns
Propagation delay time, high level to off state, $\overline{OE}$ to Y	t_{PHZ2}		02,03	2.0	7.0	ns
Propagation delay time, high level to off state, OE to Y	t_{PHZ3}		02	2.0	7.0	ns
Propagation delay time, off state to low level, $\overline{OE}$ to $\overline{Y}$	t_{PZL1}		01	2.0	10.5	ns
	t_{PZL4}		04	3.5	11.0	ns
Propagation delay time, off state to low level, $\overline{OE}$ to Y	t_{PZL2}		02,03	2.0	8.5	ns

See footnote at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C < T_C < +125°C$ unless otherwise specified	Device type	Limits		Unit
				Min	Max	
Propagation delay time, off state to low level, OE to Y	t_{PZL3}	V_{CC} = 5.0 V, C_L = 50 pF ±10%, R_L = 500Ω ±5% (see figure 4)	02	2.0	8.5	ns
Propagation delay time, off state to high level, $\overline{OE}$ to Y	t_{PZH1} t_{PZH2}		01 04	2.0 2.5	6.5 9.0	ns ns
Propagation delay time, off state to high level, $\overline{OE}$ to Y	t_{PZH2}		02,03	2.0	7.0	ns
Propagation delay time, off state to high level, OE to Y	t_{PZH3}		02	2.0	7.0	ns

1/ Not more than one output should be shorted at a time.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 9 (see MIL-M-38510, appendix E).

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7, 9,10,11	1*,2,3,7, 9
Group A test requirements (method 5005)	1,2,3,7, 9,10,11	1,2,3,7,9, 10,11
Group B test requirements (method 5005) subgroup 5	1,2,3,7, 9,10, 11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

* Indicates PDA applies to subgroup 1 (see 4.2c).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D, E, or F using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameter tests prior to burn-in are optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, 6, and 8 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical parameters shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A inspection.

c. Steady state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D, E, or F using the circuit shown on figure 3, or equivalent.

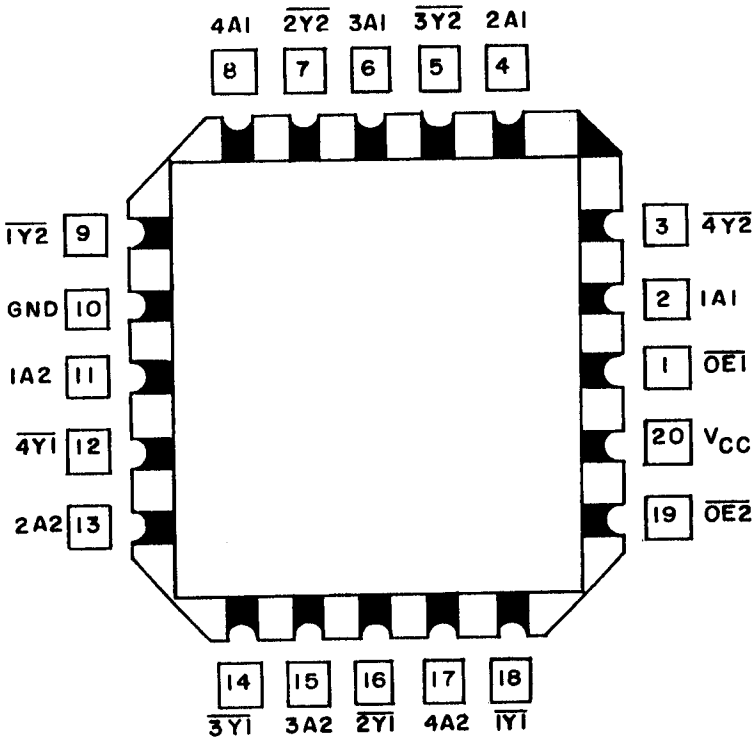
(2) T_A = +125 C minimum.

(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

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Device type 01

Cases 2 and X



Cases R and S

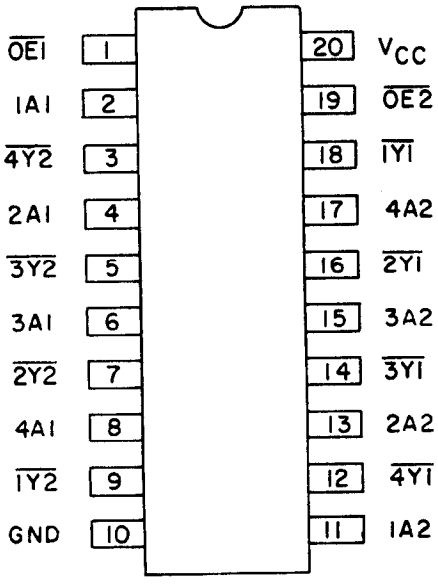
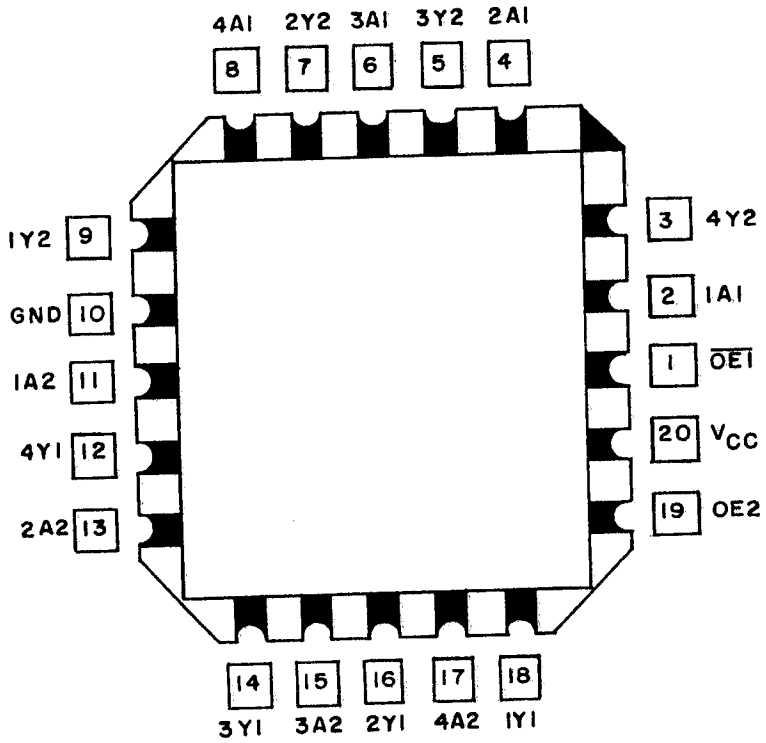


FIGURE 1. Terminal connections (top view).

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Device type 02

Cases 2 and X



Cases R and S

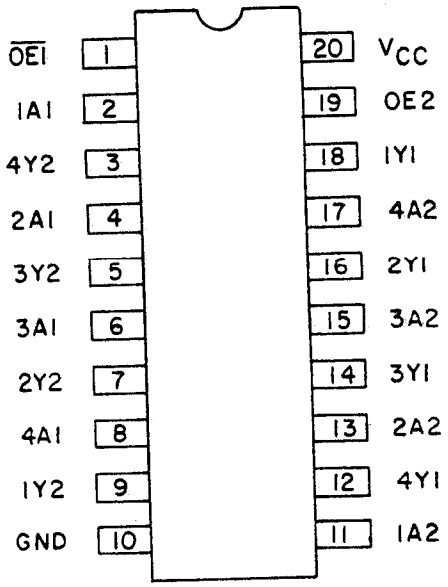
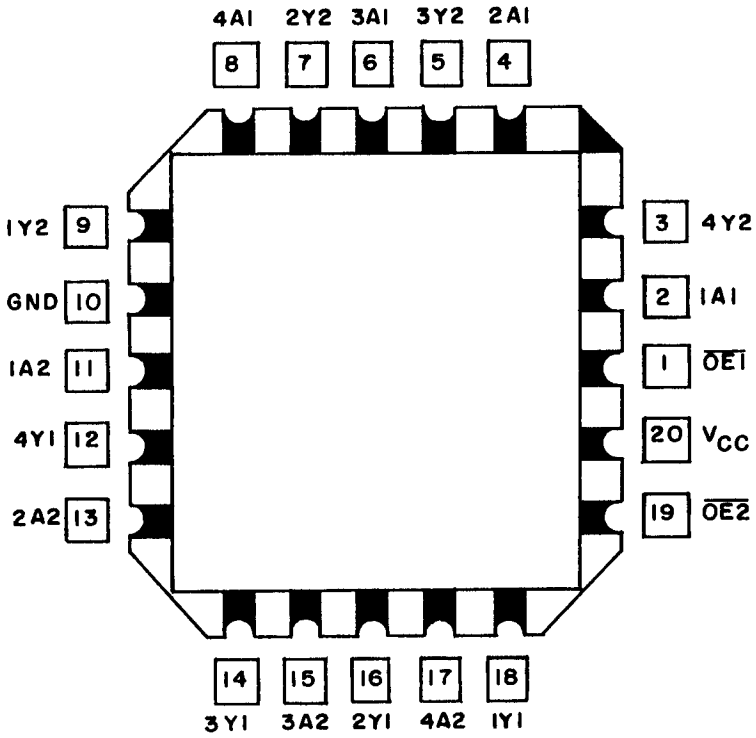


FIGURE 1. Terminal connections (top view) - Continued.

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Device type 03

Cases 2 and X



Cases R and S

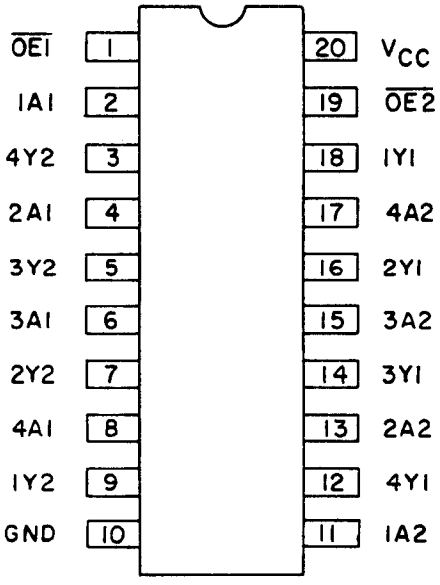
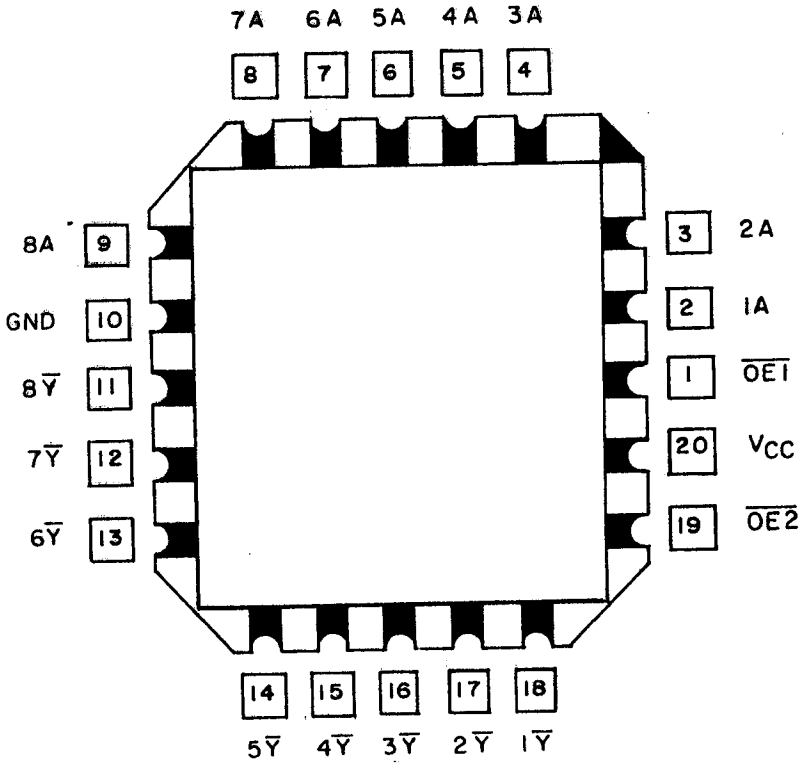


FIGURE 1. Terminal connections (top view) - Continued.

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Device type 04

Cases 2 and X



Device type 04

Case R and S

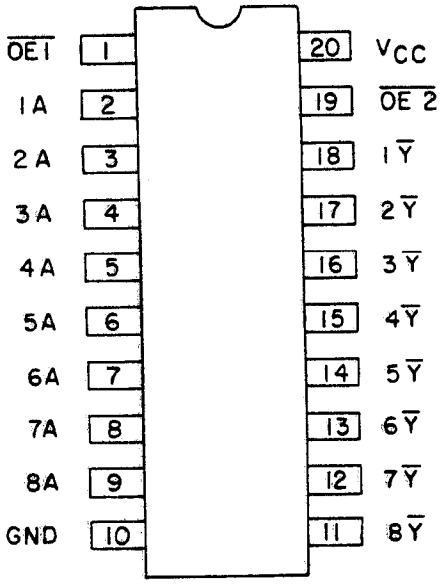
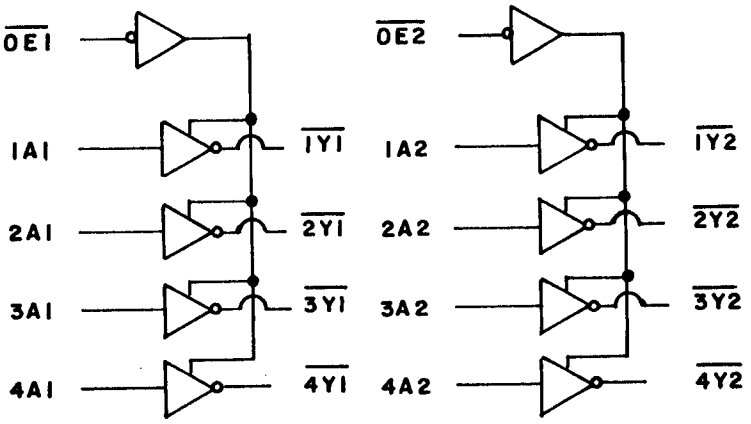


FIGURE 1. Terminal connections (top view) - Continued.

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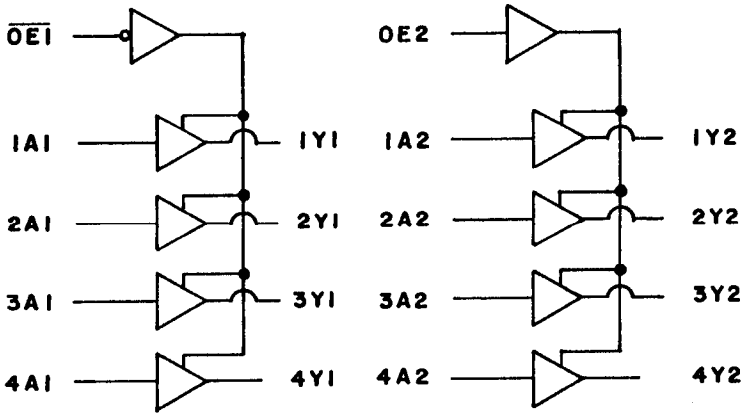
Device type 01



Inputs		Outputs
$\overline{OE1}$	A1	$\overline{Y1}$
L	L	H
L	H	L
H	X	Z

Inputs		Outputs
$\overline{OE2}$	A2	$\overline{Y2}$
L	L	H
L	H	L
H	X	X

Device type 02



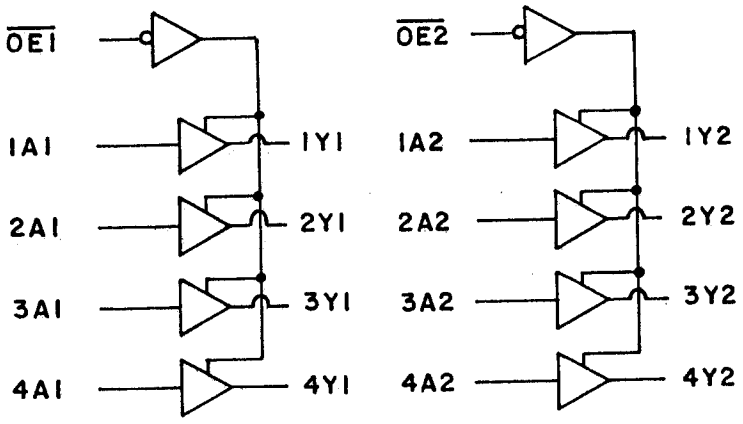
Inputs		Outputs
$\overline{OE1}$	A1	Y1
L	L	L
L	H	H
H	X	Z

Inputs		Outputs
OE2	A2	Y2
H	L	L
H	H	H
L	X	Z

FIGURE 2. Logic diagrams and truth tables.

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Device type 03



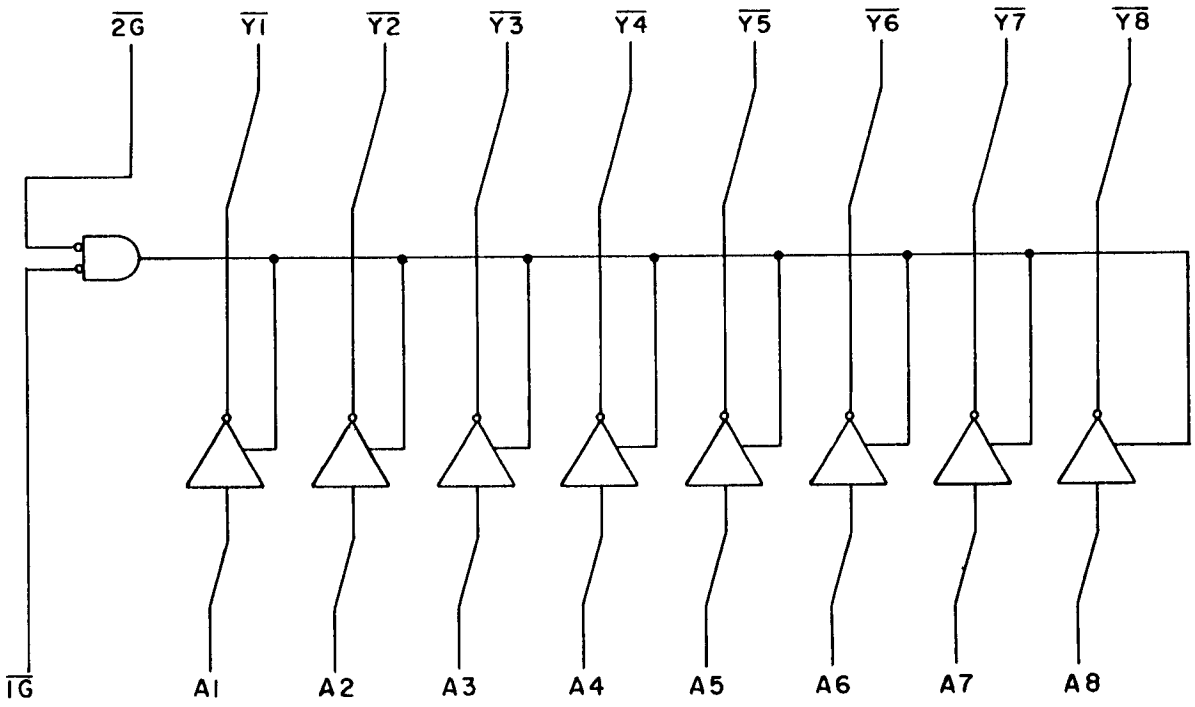
Inputs		Outputs
$\overline{OE1}$	A1	Y1
L	L	L
L	H	H
H	X	Z

Inputs		Outputs
$\overline{OE2}$	A2	Y2
L	L	L
L	H	H
H	X	Z

FIGURE 2. Logic diagrams and truth tables - Continued.

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Device type 04



Truth table

Inputs			Outputs
$\overline{OE}_1$	$\overline{OE}_2$	D	$\overline{Y}$
L	L	H	L
H	X	X	Z
X	H	X	Z
L	L	L	H

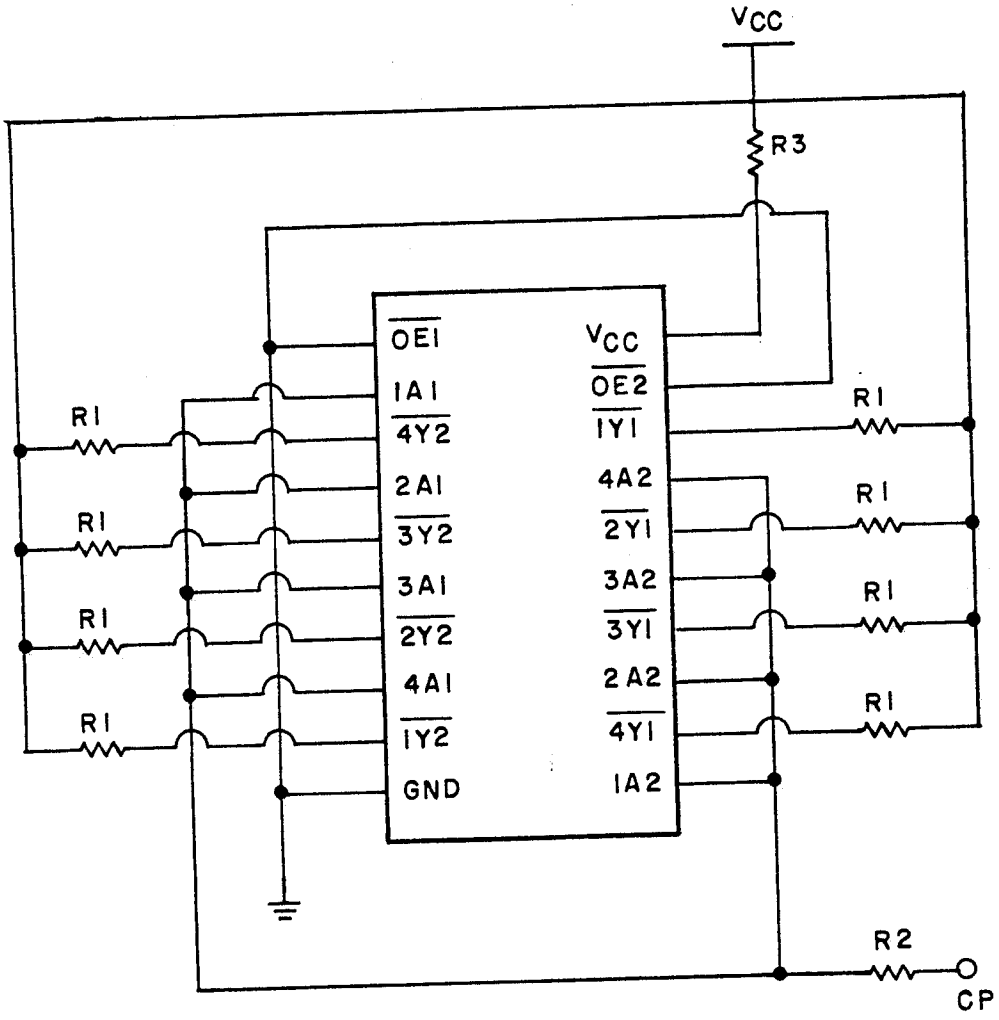
H = HIGH voltage level
L = LOW voltage level
X = Immaterial
Z = High impedance

FIGURE 2. Logic diagram and truth table - Continued.

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Device types 01 and 03 see note 4.

Cases R, S, X, and 2



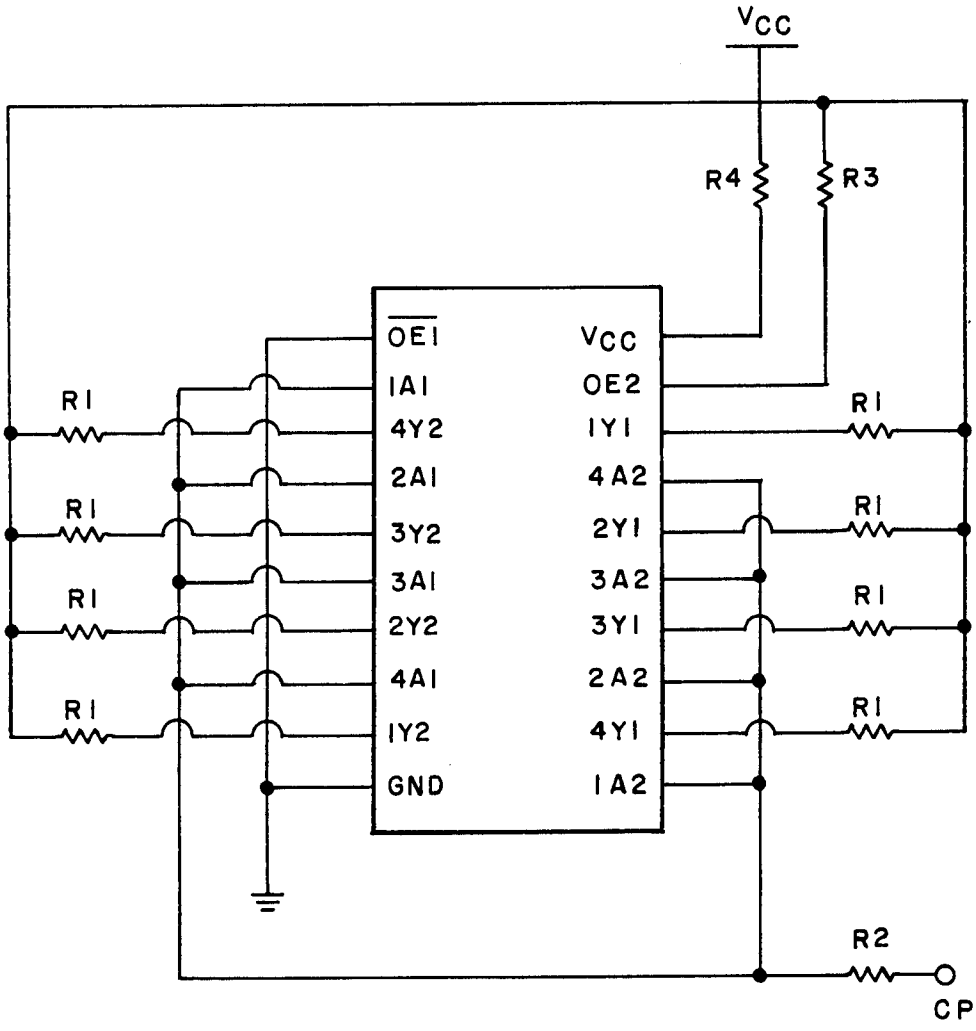
NOTES:
1. CP or $\overline{CP}$ = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 100Ω ±5%, R2 = 50Ω ±5%.
3. V_{CC} and R3 shall be chosen to insure 5.5 V minimum is present at device V_{CC} terminal.
4. Device type 03 has noninverting outputs.

FIGURE 3. Burn-in and life test circuit.

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Device type 02

Cases R, S, X, and 2

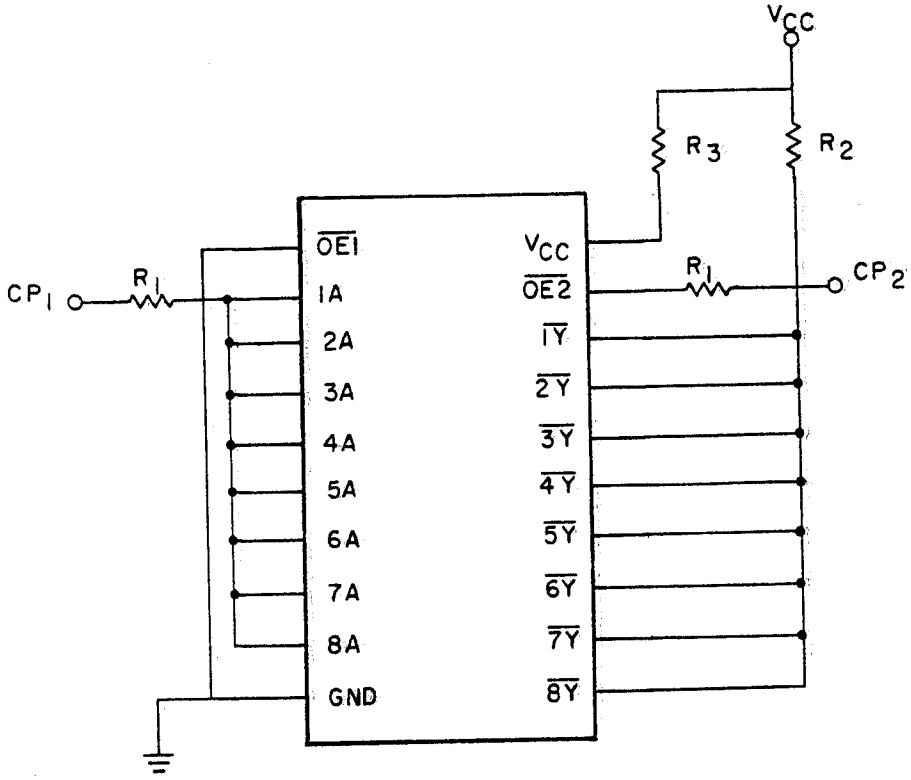


NOTES:
1. CP or $\overline{CP}$ = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 100Ω ±5%, R2 = 50Ω, ±5%, R3 = 1 kΩ ±5%.
3. V_{CC} and R4 shall be chosen to insure 5.5 V minimum is present at device V_{CC} terminal.

FIGURE 3. Burn-in and life test circuit - Continued.

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Device type 04

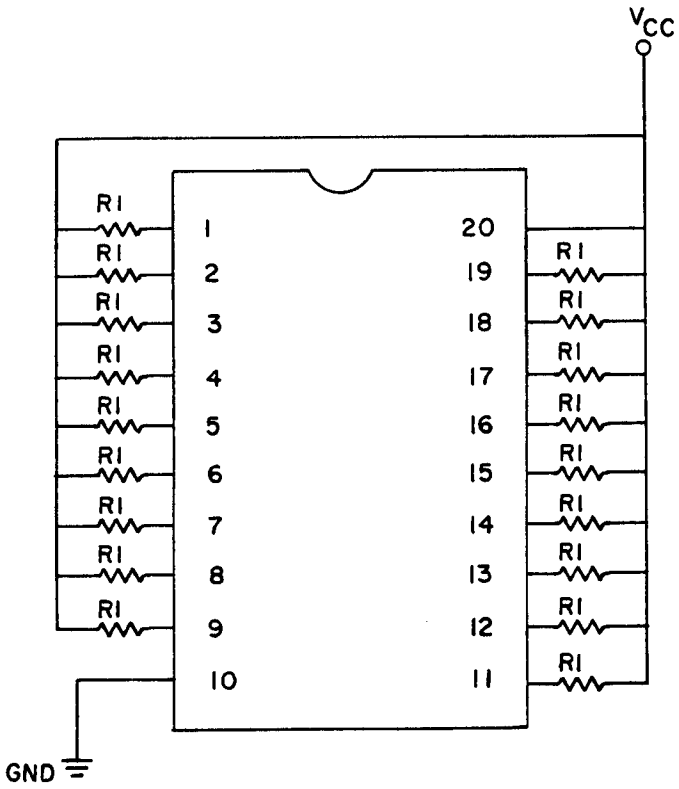


NOTES:
1. CP_1 = 100 kHz ±50% square wave; duty cycle = 50 ±15%; $CP_2 = CP_1 \div 2$; V_{IL} = -0.5 V minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 51Ω ±5%, R2 = 240Ω maximum.
3. V_{CC} and R3 shall be chosen to insure 5.5 V minimum is present at device V_{CC} terminal.

FIGURE 3. Burn-in and life circuit - Continued.

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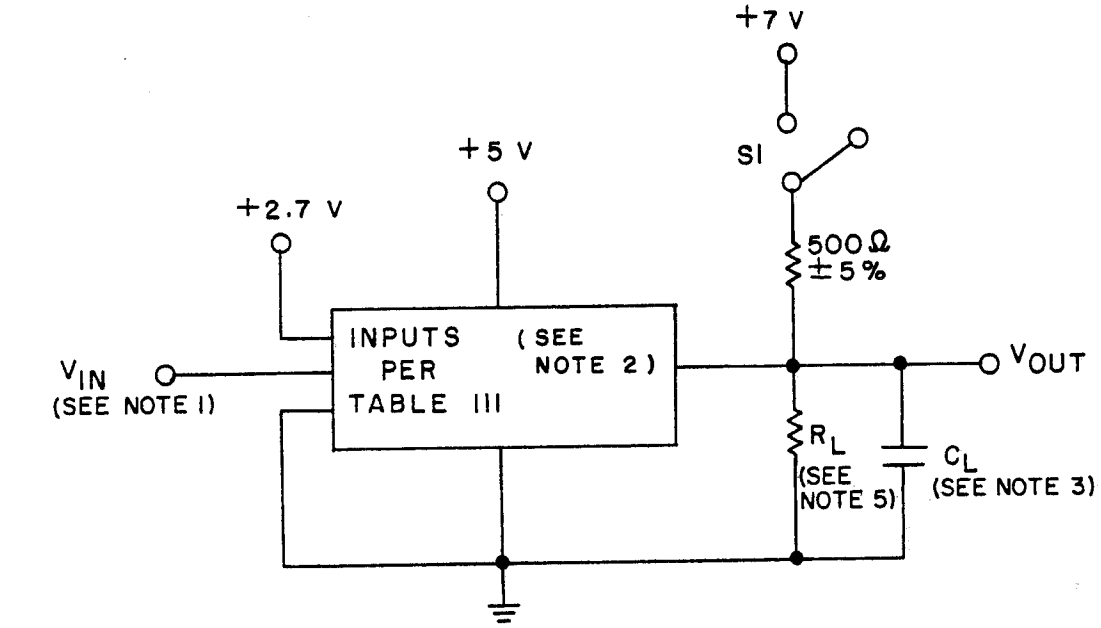
All device types

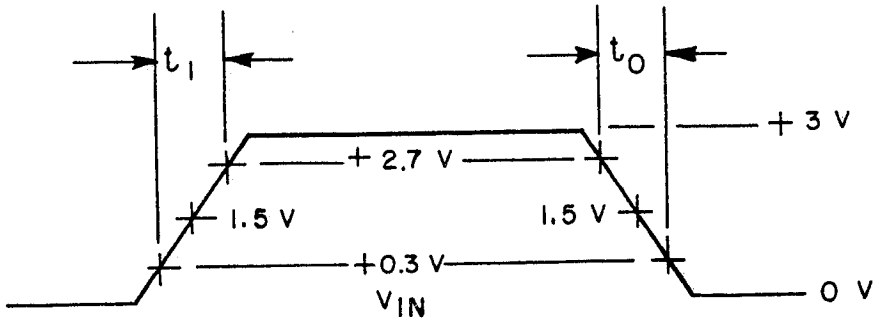


NOTES:
1. High temperature (175°C) condition F burn-in circuit only.
2. R1 = 2 kΩ.
3. V_{CC} shall be chosen to insure 5.0 V minimum is present at device V_{CC} terminal.

FIGURE 3. Burn-in and steady state life test circuit - Continued.

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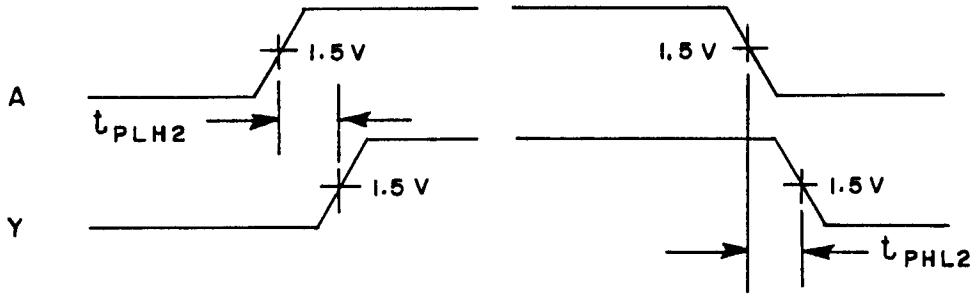
Test type	S1
t_{PLH}	Open
t_{PHL}	Open
t_{PHZ}	Open
t_{PZH}	Open
t_{PLZ}	Closed
t_{PZL}	Closed

NOTES:
1. Pulse generator has the following characteristics: $t_1 = t_0 \leq 2.5$ ns; PRR $\leq$ 1 MHz, and $Z_{OUT} \cong 50\Omega$.
2. Inputs not under test are at 2.7 V or GND as specified in table III.
3. C_L = 50 pF ±10% including scope, probe, wiring and stray capacitance, without package in test fixture.
4. Voltage measurements are to be made with respect to network ground terminal.
5. $R_L = 500\Omega \pm 5\%$.

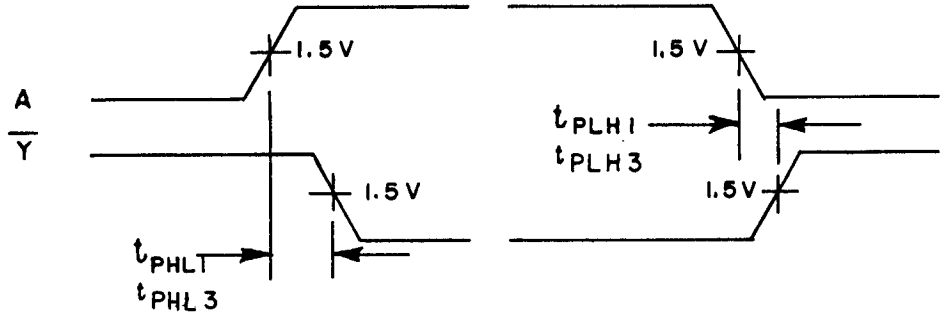
FIGURE 4. Switching time test circuit and waveforms (all device types).

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Device types 02 and 03



Device types 01 and 04



Device types 01, 02, 03, and 04

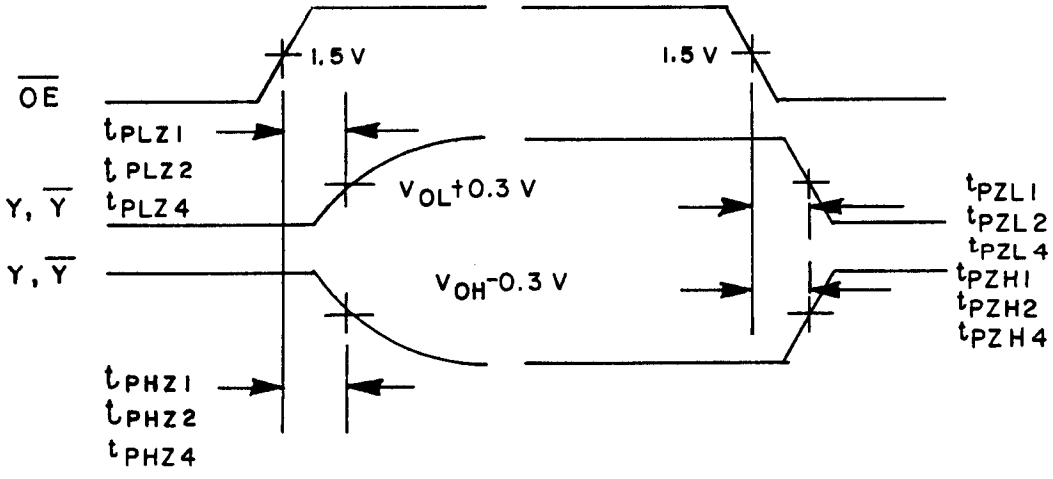


FIGURE 4. Switching time test circuit and waveforms (all device types) - Continued.

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Device type 02

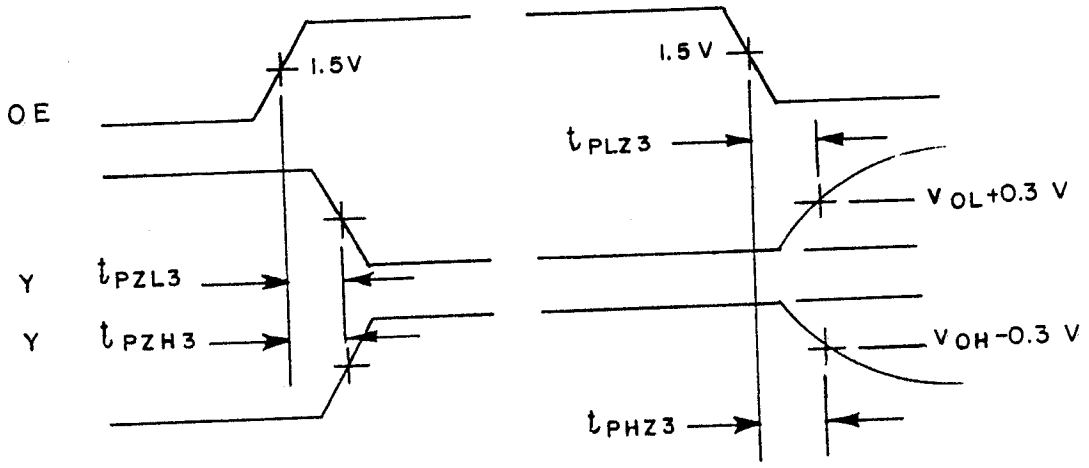


FIGURE 4. Switching time test circuit and waveforms - Continued.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases R, S, 2, X Test no.	1 $\overline{OE1}$	2 1A1	3 $\overline{4Y2}$	4 2A1	5 $\overline{3Y2}$	6 3A1	7 $\overline{2Y2}$	8 4A1	9 $\overline{1Y2}$	10 GND	11 1A2	12 $\overline{4Y1}$	13 2A2	14 $\overline{3Y1}$	15 3A2	16 $\overline{2Y1}$	17 4A2	18 $\overline{1Y1}$	19 $\overline{OE2}$	20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	V_{OH}	3006	1	0.8 V	0.8 V		0.8 V						GND						-3 mA		-3 mA		4.5 V	$\overline{1Y1}$	2.4		V
			2	"					0.8 V				"											$\overline{2Y1}$	"		"
			3	"							0.8 V		"			-3 mA								$\overline{3Y1}$	"		"
			4									-3 mA	"		-3 mA									$\overline{4Y1}$	"		"
			5										"	0.8 V								0.8 V		$\overline{1Y2}$	"		"
			6										"		-3 mA									$\overline{2Y2}$	"		"
			7					-3 mA		-3 mA			"				0.8 V			0.8 V				$\overline{3Y2}$	"		"
			8			-3 mA							"											$\overline{4Y2}$	"		"
	V_{OL}	3007	9	0.8 V	2.0 V		2.0 V						"						48 mA		48 mA		"	$\overline{1Y1}$		.55	"
			10	"					2.0 V		2.0 V		"			48 mA								$\overline{2Y1}$			"
			11	"								48 mA	"		48 mA									$\overline{3Y1}$			"
			12										"	2.0 V								0.8 V		$\overline{4Y1}$			"
			13										"		2.0 V									$\overline{1Y2}$			"
			14										"			48 mA								$\overline{2Y2}$			"
			15										"				2.0 V			2.0 V				$\overline{3Y2}$			"
			16			48 mA		48 mA		48 mA			"											$\overline{4Y2}$			"
	V_{IC}		17	-18 mA	-18 mA								"										"	$\overline{OE1}$		-1.2	"
			18				-18 mA		-18 mA		-18 mA		"											1A1			"
			19										"	-18 mA										2A1			"
			20										"		-18 mA									3A1			"
			21										"				-18 mA							4A1			"
			22										"											1A2			"
			23										"											2A2			"
			24										"											3A2			"
			25										"							-18 mA				4A2			"
			26										"											$\overline{OE2}$			"
	I_{IH1}	3010	27	2.7 V	2.7 V		2.7 V						"										5.5 V	$\overline{OE1}$		20	µA
			28						2.7 V		2.7 V		"										"	1A1			"
			29										"	2.7 V										2A1			"
			30										"		2.7 V									3A1			"
			31										"				2.7 V							4A1			"
			32										"											1A2			"
			33										"											2A2			"
			34										"											3A2			"
			35										"							2.7 V				4A2			"
			36										"											$\overline{OE2}$			"
	I_{IH2}	3010	37	7.0 V	7.0 V		7.0 V						"										5.5 V	$\overline{OE1}$		100	"
			38						7.0 V		7.0 V		"										"	1A1			"
			39										"	7.0 V										2A1			"
			40										"		7.0 V									3A1			"
			41										"				7.0 V							4A1			"
			42										"											1A2			"
			43										"											2A2			"
			44										"											3A2			"
			45										"							7.0 V				4A2			"
			46										"											$\overline{OE2}$			"
	I_{IL1}	3009	47	0.5 V	0.5 V		0.5 V						"										"	$\overline{OE1}$	1/	1/	1/
			48						0.5 V		0.5 V		"										"	1A1	"	"	"
			49										"	0.5 V										2A1	"	"	"
			50										"		0.5 V									3A1	"	"	"
			51										"				0.5 V							4A1	"	"	"
			52										"											1A2	"	"	"
			53										"											2A2	"	"	"
			54										"											3A2	"	"	"
			55										"							0.5 V				4A2	"	"	"
			56										"											$\overline{OE2}$	"	"	"

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≤2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases R, S, 2, X Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Limits Min	Limits Max	Unit
				$\overline{OE1}$	1A1	$\overline{4Y2}$	2A1	$\overline{3Y2}$	3A1	$\overline{2Y2}$	4A1	$\overline{1Y2}$	GND	1A2	$\overline{4Y1}$	2A2	$\overline{3Y1}$	3A2	$\overline{2Y1}$	4A2	$\overline{1Y1}$	$\overline{OE2}$	V_{CC}				
1 T_C = +25°C	I_{OZH}		57	2.0 V	4.5 V		4.5 V		4.5 V		4.5 V	2.4 V	GND	4.5 V	2.4 V		2.4 V		2.4 V		2.4 V		5.5 V	$\overline{1Y1}$		50	µA
			58	"									"										"	$\overline{2Y1}$		"	
			59	"									"										"	$\overline{3Y1}$		"	
			60	"									"										"	$\overline{4Y1}$		"	
			61	"									"										"	$\overline{1Y2}$		"	
			62	"									"										"	$\overline{2Y2}$		"	
			63	"									"										"	$\overline{3Y2}$		"	
			64	"		2.4 V		2.4 V		2.4 V			"			4.5 V		4.5 V		4.5 V		2.0 V	"	$\overline{4Y2}$		"	
	I_{OZL}		65	2.0 V	0.0 V		0.0 V		0.0 V		0.0 V	0.5 V	"	0.0 V	0.5 V		0.5 V		0.5 V		0.5 V		"	$\overline{1Y1}$		-50	"
			66	"									"										"	$\overline{2Y1}$		"	
			67	"									"										"	$\overline{3Y1}$		"	
			68	"									"										"	$\overline{4Y1}$		"	
			69	"									"										"	$\overline{1Y2}$		"	
			70	"									"										"	$\overline{2Y2}$		"	
			71	"									"										"	$\overline{3Y2}$		"	
			72	"		0.5 V		0.5 V		0.5 V			"			0.0 V		0.0 V		0.0 V		2.0 V	"	$\overline{4Y2}$		"	
	I_{OS}	3011	73	0.0 V	0.0 V		0.0 V		0.0 V		0.0 V	0.0 V	"	0.0 V	0.0 V		0.0 V		0.0 V		0.0 V		"	$\overline{1Y1}$	-100	-225 2/	mA
			74	"									"										"	$\overline{2Y1}$	"	"	
			75	"									"										"	$\overline{3Y1}$	"	"	
			76	"									"										"	$\overline{4Y1}$	"	"	
			77	"									"										"	$\overline{1Y2}$	"	"	
			78	"									"										"	$\overline{2Y2}$	"	"	
			79	"									"										"	$\overline{3Y2}$	"	"	
			80	"		0.0 V		0.0 V		0.0 V			"			0.0 V		0.0 V		0.0 V		0.0 V	"	$\overline{4Y2}$	"	"	
	I_{OD}		81	GND	5.5 V		5.5 V		5.5 V		5.5 V	2.5 V	"	5.5 V	2.5 V		2.5 V		2.5 V		2.5 V		4.5 V	$\overline{1Y1}$	65		"
			82	"									"										"	$\overline{2Y1}$	"		
			83	"									"										"	$\overline{3Y1}$	"		
			84	"									"										"	$\overline{4Y1}$	"		
			85	"									"										"	$\overline{1Y2}$	"		
			86	"									"										"	$\overline{2Y2}$	"		
			87	"									"										"	$\overline{3Y2}$	"		
			88	"		2.5 V		2.5 V		2.5 V			"			5.5 V		5.5 V		5.5 V		GND	"	$\overline{4Y2}$	"		
	I_{CCH}	3005	89	0.0 V	0.0 V		0.0 V		0.0 V		0.0 V		"	0.0 V		0.0 V		0.0 V		0.0 V		0.0 V	5.5 V	V_{CC}		35	"
	I_{CCL}		90	0.0 V	4.5 V		4.5 V		4.5 V		4.5 V		"	4.5 V		4.5 V		4.5 V		4.5 V		0.0 V	"	V_{CC}		75	"
	I_{CCZ}		91	4.5 V									"									4.5 V	"	V_{CC}		75	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																										
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																										
7 3/ T_C = +25°C	Functional tests	3014	92	B	B	A	H	B	H	B	H	B	H	GND	B	H	B	H	B	H	B	H	B	4/	All outputs			
		3014	93	A		L	A		L	A	L	A	L	"	A	L	A	L	A	L	A	L	B	4/	All outputs			
9 T_C = +25°C	t_{PLH1}	3003 (Fig. 4)	94	0.0 V	IN		IN			IN		IN		"	IN			OUT		OUT		OUT		5.0 V	1A1 to $\overline{1Y1}$	1.5	7.0	ns
			95	"										"										"	2A1 to $\overline{2Y1}$	"	"	
			96	"										"										"	3A1 to $\overline{3Y1}$	"	"	
			97	"										"										"	4A1 to $\overline{4Y1}$	"	"	
			98	"										"										"	1A2 to $\overline{1Y2}$	"	"	
			99	"										"										"	2A2 to $\overline{2Y2}$	"	"	
			100	"										"										"	3A2 to $\overline{3Y2}$	"	"	
			101	"		OUT		OUT		OUT				"									0.0 V	"	4A2 to $\overline{4Y2}$	"	"	

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≤2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases R, S, 2, X	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Limits		Unit
			Test no.	$\overline{OE1}$	1A1	$\overline{4Y2}$	2A1	$\overline{3Y2}$	3A1	$\overline{2Y2}$	4A1	$\overline{1Y2}$	GND	1A2	$\overline{4Y1}$	2A2	$\overline{3Y1}$	3A2	$\overline{2Y1}$	4A2	$\overline{1Y1}$	$\overline{OE2}$	V_{CC}		Min	Max	
9 T_C = +25°C	t_{PHL1}	3003 (Fig. 4)	102	0.0 V	IN		IN		IN		IN		GND						OUT		OUT		5.0 V	1A1 to $\overline{1Y1}$	1.5	4.7	ns
			103	"									"										"	2A1 to $\overline{2Y1}$	"	"	"
			104	"									"				OUT						"	3A1 to $\overline{3Y1}$	"	"	"
			105	"									"		OUT								"	4A1 to $\overline{4Y1}$	"	"	"
			106									OUT	"	IN								0.0 V	"	1A2 to $\overline{1Y2}$	"	"	"
			107							OUT			"			IN							"	2A2 to $\overline{2Y2}$	"	"	"
			108					OUT					"					IN					"	3A2 to $\overline{3Y2}$	"	"	"
			109			OUT							"							IN			"	4A2 to $\overline{4Y2}$	"	"	"
	t_{PHZ1}	3003 (Fig. 4)	110	IN	0.0 V		0.0 V		0.0 V		0.0 V		"						OUT		OUT		"	$\overline{OE1}$ to $\overline{1Y1}$	2.0	5.3	"
			111	"									"				OUT						"	$\overline{OE1}$ to $\overline{2Y1}$	"	"	"
			112	"									"		OUT								"	$\overline{OE1}$ to $\overline{3Y1}$	"	"	"
			113	"									"										"	$\overline{OE1}$ to $\overline{4Y1}$	"	"	"
			114										"	0.0 V		0.0 V						IN	"	$\overline{OE2}$ to $\overline{1Y2}$	"	"	"
			115							OUT			"					0.0 V					"	$\overline{OE2}$ to $\overline{2Y2}$	"	"	"
			116					OUT					"							0.0 V			"	$\overline{OE2}$ to $\overline{3Y2}$	"	"	"
			117			OUT							"										"	$\overline{OE2}$ to $\overline{4Y2}$	"	"	"
	t_{PLZ1}	3003 (Fig. 4)	118	IN	2.7 V		2.7 V		2.7 V		2.7 V		"						OUT		OUT		"	$\overline{OE1}$ to $\overline{1Y1}$	"	10.0	"
			119	"									"				OUT						"	$\overline{OE1}$ to $\overline{2Y1}$	"	"	"
			120	"									"		OUT								"	$\overline{OE1}$ to $\overline{3Y1}$	"	"	"
			121	"									"										"	$\overline{OE1}$ to $\overline{4Y1}$	"	"	"
			122										"	2.7 V		2.7 V						IN	"	$\overline{OE2}$ to $\overline{1Y2}$	"	"	"
			123							OUT			"					2.7 V					"	$\overline{OE2}$ to $\overline{2Y2}$	"	"	"
			124					OUT					"							2.7 V			"	$\overline{OE2}$ to $\overline{3Y2}$	"	"	"
			125			OUT							"										"	$\overline{OE2}$ to $\overline{4Y2}$	"	"	"
	t_{PZH1}	3003 (Fig. 4)	126	IN	0.0 V		0.0 V		0.0 V		0.0 V		"						OUT		OUT		"	$\overline{OE1}$ to $\overline{1Y1}$	"	5.5	"
			127	"									"				OUT						"	$\overline{OE1}$ to $\overline{2Y1}$	"	"	"
			128	"									"		OUT								"	$\overline{OE1}$ to $\overline{3Y1}$	"	"	"
			129	"									"										"	$\overline{OE1}$ to $\overline{4Y1}$	"	"	"
			130										"	0.0 V		0.0 V						IN	"	$\overline{OE2}$ to $\overline{1Y2}$	"	"	"
			131							OUT			"					0.0 V					"	$\overline{OE2}$ to $\overline{2Y2}$	"	"	"
			132					OUT					"							0.0 V			"	$\overline{OE2}$ to $\overline{3Y2}$	"	"	"
			133			OUT							"										"	$\overline{OE2}$ to $\overline{4Y2}$	"	"	"
	t_{PZL1}	3003 (Fig. 4)	134	IN	2.7 V		2.7 V		2.7 V		2.7 V		"						OUT		OUT		"	$\overline{OE1}$ to $\overline{1Y1}$	"	9.0	"
			135	"									"				OUT						"	$\overline{OE1}$ to $\overline{2Y1}$	"	"	"
			136	"									"		OUT								"	$\overline{OE1}$ to $\overline{3Y1}$	"	"	"
			137	"									"										"	$\overline{OE1}$ to $\overline{4Y1}$	"	"	"
			138										"	2.7 V		2.7 V						IN	"	$\overline{OE2}$ to $\overline{1Y2}$	"	"	"
			139							OUT			"					2.7 V					"	$\overline{OE2}$ to $\overline{2Y2}$	"	"	"
			140					OUT					"							2.7 V			"	$\overline{OE2}$ to $\overline{3Y2}$	"	"	"
			141			OUT							"										"	$\overline{OE2}$ to $\overline{4Y2}$	"	"	"
10 T_C = +125°C	t_{PLH1}	Same tests and terminal conditions as for subgroup 9, except T_C = +125°.																							1.0	9.0	"
	t_{PHL1}																								1.0	6.0	"
	t_{PHZ1}																								2.0	6.5	"
	t_{PLZ1}																								"	12.5	"
	t_{PZH1}																								"	6.5	"
	t_{PZL1}																								"	10.5	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																										

1/ I_{IL} limits shall be as follows:

Test	Min/Max limits in (mA) Circuits A	B	C
I_{IL1}	-.42/-1.0	-.03/-1.0	-.03/-1.0

2/ I_{OS} limits (mA) min/max value for CKT B, -100/-325.

3/ A = 2.4 V minimum; B = 0.55 V or GND, H ≥ 1.5, L ≤ 1.5 V.

4/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

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TABLE III. Group A inspection for device type 02.

Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases R, S, 2, X Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Limits Min	Limits Max	Unit
				OE1	1A1	4Y2	2A1	3Y2	3A1	2Y2	4A1	1Y2	GND	1A2	4Y1	2A2	3Y1	3A2	2Y1	4A2	1Y1	OE2	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1 2 3 4 5 6 7 8	0.8 V " "	2.0 V	-3 mA	2.0 V	-3 mA	2.0 V	-3 mA	2.0 V	-3 mA	GND " " " " " "	2.0 V	-3 mA	2.0 V	-3 mA	2.0 V	-3 mA	2.0 V	-3 mA	2.0 V " "	4.5 V " " " " " "	1Y1 2Y1 3Y1 4Y1 1Y2 2Y2 3Y2 4Y2	2.4 " " " " " "		V " " " " " "
	V_{OL}	3007	9 10 11 12 13 14 15 16	0.8 V " "	0.8 V	48 mA	0.8 V	48 mA	0.8 V	48 mA	0.8 V	48 mA	" " " " " " " "	0.8 V	48 mA	0.8 V	48 mA	0.8 V	48 mA	0.8 V	48 mA	2.0 V " "	" " " " " " " "	1Y1 2Y1 3Y1 4Y1 1Y2 2Y2 3Y2 4Y2		.55 " " " " " "	" " " " " " " "
	V_{IC}		17 18 19 20 21 22 23 24 25 26	-18 mA	-18 mA		-18 mA		-18 mA		-18 mA		" " " " " " " " " "	-18 mA		-18 mA		-18 mA		-18 mA		-18 mA	" " " " " " " " " "	OE1 1A1 2A1 3A1 4A1 1A2 2A2 3A2 4A2 OE2		-1.2 " " " " " "	" " " " " " "
	I_{IH1}	3010	27 28 29 30 31 32 33 34 35 36	2.7 V	2.7 V		2.7 V		2.7 V		2.7 V		" " " " " " " " " "	2.7 V		2.7 V		2.7 V		2.7 V		2.7 V	5.5 V " " " " " " " " "	OE1 1A1 2A1 3A1 4A1 1A2 2A2 3A2 4A2 OE2		20 " " " " " "	μA " " " " " "
	I_{IH2}	3010	37 38 39 40 41 42 43 44 45 46	7.0 V	7.0 V		7.0 V		7.0 V		7.0 V		" " " " " " " " " "	7.0 V		7.0 V		7.0 V		7.0 V		7.0 V	" " " " " " " " " "	OE1 1A1 2A1 3A1 4A1 1A2 2A2 3A2 4A2 OE2		100 " " " " " "	" " " " " " "
	I_{IL1}	3009	47 48 49 50 51 52 53 54		0.5 V		0.5 V		0.5 V		0.5 V		" " " " " " " "	0.5 V		0.5 V		0.5 V		0.5 V			" " " " " " " "	1A1 2A1 3A1 4A1 1A2 2A2 3A2 4A2	1/ " " " " " "	1/ " " " " " "	1/ " " " " " "
	I_{IL2} I_{IL2}	3009 3009	55 56	0.5 V									" "									0.5 V	" "	OE1 OE2	" "	" "	" "

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases R, S, 2, X Test no.	1 $\overline{OE1}$	2 1A1	3 4Y2	4 2A1	5 3Y2	6 3A1	7 2Y2	8 4A1	9 1Y2	10 GND	11 1A2	12 4Y1	13 2A2	14 3Y1	15 3A2	16 2Y1	17 4A2	18 1Y1	19 $\overline{OE2}$	20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	I_{OZH}		57	2.0 V	0.0 V		0.0 V		0.0 V				GND						2.4 V		2.4 V		5.5 V	1Y1		50	µA
			58	"									"										"	2Y1		"	"
			59	"									"										"	3Y1		"	"
			60	"							0.0 V		"		2.4 V								"	4Y1		"	"
			61									2.4 V	"	0.0 V									"	1Y2		"	"
			62							2.4 V			"								0.8 V		"	2Y2		"	"
			63					2.4 V					"				0.0 V				"		"	3Y2		"	"
			64			2.4 V							"						0.0 V		"		"	4Y2		"	"
	I_{OZL}		65	2.0 V	4.5 V								"										"	1Y1		-50	"
			66	"			4.5 V						"										"	2Y1		"	"
			67	"					4.5 V				"										"	3Y1		"	"
			68								4.5 V		"		0.5 V								"	4Y1		"	"
			69									0.5 V	"	4.5 V									"	1Y2		"	"
			70					0.5 V					"				4.5 V				0.8 V		"	2Y2		"	"
			71							0.5 V			"		4.5 V						"		"	3Y2		"	"
			72			0.5 V							"						4.5 V				"	4Y2		"	"
	I_{OS}	3011	73	0.0 V	4.5 V								"							0.0 V			"	1Y1	-100	-225 2/	mA
			74	"			4.5 V						"										"	2Y1	"	"	"
			75	"					4.5 V				"										"	3Y1	"	"	"
			76	"							4.5 V		"		0.0 V								"	4Y1	"	"	"
			77									0.0 V	"	4.5 V									"	1Y2	"	"	"
			78							0.0 V			"		4.5 V						4.5 V		"	2Y2	"	"	"
			79					0.0 V					"				4.5 V				"		"	3Y2	"	"	"
			80			0.0 V							"						4.5 V				"	4Y2	"	"	"
	I_{OD}		81	GND	GND								"										4.5 V	1Y1	65		"
			82	"			GND						"										"	2Y1	"		"
			83	"					GND				"										"	3Y1	"		"
			84								GND		"		2.5 V								"	4Y1	"		"
			85									2.5 V	"	GND							5.5 V		"	1Y2	"		"
			86							2.5 V			"		GND						"		"	2Y2	"		"
			87					2.5 V					"						GND		"		"	3Y2	"		"
			88			2.5 V							"										"	4Y2	"		"
	I_{CCH}	3005	89	0.0 V	4.5 V		4.5 V		4.5 V		4.5 V		"	4.5 V		4.5 V		4.5 V		4.5 V		4.5 V	5.5 V	V_{CC}		60	"
	I_{CCL}	3005	90	0.0 V	0.0 V		0.0 V		0.0 V		0.0 V		"	0.0 V		0.0 V		0.0 V		0.0 V		4.5 V	"	V_{CC}		90	"
	I_{CCZ}	3005	91	4.5 V									"								0.0 V	"	"	V_{CC}		90	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																										
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																										
7 3/ T_C = +25°C	Functional tests	3014	92	B	B	L	B	L	B	L	B	L	GND	B	L	B	L	B	L	B	L	A	4/	All outputs			
		3014	93	B	A	H	A	H	A	H	A	H	"	A	H	A	H	A	H	A	H	A	4/	All outputs			
9 T_C = +25°C	t_{PLH2}	3003 (Fig. 4)	94	0.0 V	IN		IN		IN		IN		"	IN			OUT		OUT		OUT		5.0 V	1A1 to 1Y1	1.5	5.2	ns
			95	"									"		OUT								"	2A1 to 2Y1	"	"	"
			96	"									"				OUT						"	3A1 to 3Y1	"	"	"
			97										"			IN							"	4A1 to 4Y1	"	"	"
			98										"									2.7 V	"	1A2 to 1Y2	"	"	"
			99										"										"	2A2 to 2Y2	"	"	"
			100					OUT					"					IN					"	3A2 to 3Y2	"	"	"
			101			OUT				OUT		OUT	"										"	4A2 to 4Y2	"	"	"

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≤2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases R, S, 2, X Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Limits Min	Limits Max	Unit
				$\overline{OE1}$	1A1	4Y2	2A1	3Y2	3A1	2Y2	4A1	1Y2	GND	1A2	4Y1	2A2	3Y1	3A2	2Y1	4A2	1Y1	OE2	V_{CC}				
9 T_C = +25°C	t_{PHL2}	3003 (Fig. 4)	102 103 104 105 106 107 108 109	0.0 V " "	IN	OUT	IN	OUT	IN	OUT	IN	OUT	GND " " " " " "	IN	OUT	IN	OUT	IN	OUT	IN	OUT	2.7 V " "	5.0 V " " " " " "	1A1 to 1Y1 2A1 to 2Y1 3A1 to 3Y1 4A1 to 4Y1 1A2 to 1Y2 2A2 to 2Y2 3A2 to 3Y2 4A2 to 4Y2	1.5 " " " " " " "	5.2 " " " " " " "	ns " " " " " " "
	t_{PHZ2}	3003 (Fig. 4)	110 111 112 113	IN " " "	2.7 V		2.7 V		2.7 V		2.7 V		" " " "		OUT		OUT		OUT		OUT		" " " "	$\overline{OE1}$ to 1Y1 $\overline{OE1}$ to 2Y1 $\overline{OE1}$ to 3Y1 $\overline{OE1}$ to 4Y1	2.0 " "	6.0 " "	" " "
	t_{PHZ3}	3003 (Fig. 4)	114 115 116 117			OUT		OUT		OUT		OUT	" " " "	2.7 V		2.7 V		2.7 V		2.7 V		IN " "	" " " "	OE2 to 1Y2 OE2 to 2Y2 OE2 to 3Y2 OE2 to 4Y2	" " "	" " "	" " "
	t_{PLZ2}	3003 (Fig. 4)	118 119 120 121	IN " "	0.0 V		0.0 V		0.0 V		0.0 V		" " " "		OUT		OUT		OUT		OUT		" " " "	$\overline{OE1}$ to 1Y1 $\overline{OE1}$ to 2Y1 $\overline{OE1}$ to 3Y1 $\overline{OE1}$ to 4Y1	" " "	" " "	" " "
	t_{PLZ3}	3003 (Fig. 4)	122 123 124 125			OUT		OUT		OUT		OUT	" " " "	0.0 V		0.0 V		0.0 V		0.0 V		IN " "	" " " "	OE2 to 1Y2 OE2 to 2Y2 OE2 to 3Y2 OE2 to 4Y2	" " "	" " "	" " "
	t_{PZH2}	3003 (Fig. 4)	126 127 128 129	IN " "	2.7 V		2.7 V		2.7 V		2.7 V		" " " "		OUT		OUT		OUT		OUT		" " " "	$\overline{OE1}$ to 1Y1 $\overline{OE1}$ to 2Y1 $\overline{OE1}$ to 3Y1 $\overline{OE1}$ to 4Y1	" " "	5.7 " "	" " "
	t_{PZH3}	3003 (Fig. 4)	130 131 132 133			OUT		OUT		OUT		OUT	" " " "	2.7 V		2.7 V		2.7 V		2.7 V		IN " "	" " " "	OE2 to 1Y2 OE2 to 2Y2 OE2 to 3Y2 OE2 to 4Y2	" " "	" " "	" " "
	t_{PZL2}	3003 (Fig. 4)	134 135 136 137	IN " "	0.0 V		0.0 V		0.0 V		0.0 V		" " " "		OUT		OUT		OUT		OUT		" " " "	$\overline{OE1}$ to 1Y1 $\overline{OE1}$ to 2Y1 $\overline{OE1}$ to 3Y1 $\overline{OE1}$ to 4Y1	" " "	7.0 " "	" " "
	t_{PZL3}	3003 (Fig. 4)	138 139 140 141			OUT		OUT		OUT		OUT	" " " "	0.0 V		0.0 V		0.0 V		0.0 V		IN " "	" " " "	OE2 to 1Y2 OE2 to 2Y2 OE2 to 3Y2 OE2 to 4Y2	" " "	" " "	" " "

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≤2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases R, S, 2, X Test no.	1 $\overline{OE1}$	2 1A1	3 4Y2	4 2A1	5 3Y2	6 3A1	7 2Y2	8 4A1	9 1Y2	10 GND	11 1A2	12 4Y1	13 2A2	14 3Y1	15 3A2	16 2Y1	17 4A2	18 1Y1	19 OE2	20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
10	t_{PLH2}	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C.																						1.0	6.5	ns
	t_{PHL2}																							1.0	7.0	"
	t_{PHZ2}																							2.0	7.0	"
	t_{PHZ3}																							"	7.0	"
	t_{PLZ2}																							"	7.5	"
	t_{PLZ3}																							"	7.5	"
	t_{PZH2}																							"	7.0	"
	t_{PZH3}																							"	7.0	"
	t_{PZL2}																							"	8.5	"
	t_{PZL3}																							"	8.5	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																									

1/ I_{IL} limits shall be as follows:

Test	Min/Max limits in (mA) Circuits A	B	C
I_{IL1}	-.67/-1.6	-.03/-1.6	-.03/-1.6
I_{IL2}	-.42/-1.0	-.03/-1.0	-.06/-1.0

2/ I_{OS} limits (mA) min/max values for CKT B, -100/-325.

3/ A = 2.4 V minimum; B = 5.5 V or GND, H ≥1.5 V, L ≤ 1.5 V.

4/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases R, S, 2, X Test no.	1 $\overline{OE1}$	2 1A1	3 4Y2	4 2A1	5 3Y2	6 3A1	7 2Y2	8 4A1	9 1Y2	10 GND	11 1A2	12 4Y1	13 2A2	14 3Y1	15 3A2	16 2Y1	17 4A2	18 1Y1	19 $\overline{OE2}$	20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = 25°C	V_{OH}	3006	1	0.8 V	2.0 V		2.0 V		2.0 V		2.0 V		GND	2.0 V		2.0 V		2.0 V		2.0 V		0.8 V	4.5 V	1Y1	2.4		V
			2	"									"		-3 mA		-3 mA		-3 mA		-3 mA	"	"	2Y1	"		"
			3	"									"									"	"	3Y1	"		"
			4										"									"	"	4Y1	"		"
			5										"									"	"	1Y2	"		"
			6										"									"	"	2Y2	"		"
			7										"									"	"	3Y2	"		"
			8			-3 mA		-3 mA		-3 mA		-3 mA	"									"	"	4Y2	"		"
	V_{OL}	3007	9	0.8 V	0.8 V		0.8 V		0.8 V		0.8 V		"	0.8 V		0.8 V		0.8 V		0.8 V		0.8 V	"	1Y1		.55	"
			10	"									"		48 mA		48 mA		48 mA		48 mA	"	"	2Y1	"	"	"
			11	"									"									"	"	3Y1	"	"	"
			12										"									"	"	4Y1	"	"	"
			13										"									"	"	1Y2	"	"	"
			14										"									"	"	2Y2	"	"	"
			15										"									"	"	3Y2	"	"	"
			16			48 mA		48 mA		48 mA		48 mA	"									"	"	4Y2	"	"	"
	V_{IC}		17	-18 mA									"									"	"	$\overline{OE1}$		-1.2	"
			18		-18 mA		-18 mA		-18 mA		-18 mA		"	-18 mA		-18 mA		-18 mA		-18 mA		-18 mA	"	1A1	"	"	"
			19										"									"	"	2A1	"	"	"
			20										"									"	"	3A1	"	"	"
			21										"									"	"	4A1	"	"	"
			22										"									"	"	1A2	"	"	"
			23										"									"	"	2A2	"	"	"
			24										"									"	"	3A2	"	"	"
			25										"									"	"	4A2	"	"	"
			26										"									"	"	$\overline{OE2}$	"	"	"
	I_{IH1}	3010	27	2.7 V									"									"	"	$\overline{OE1}$		20	µA
			28		2.7 V		2.7 V		2.7 V		2.7 V		"	2.7 V		2.7 V		2.7 V		2.7 V		2.7 V	"	1A1	"	"	"
			29										"									"	"	2A1	"	"	"
			30										"									"	"	3A1	"	"	"
			31										"									"	"	4A1	"	"	"
			32										"									"	"	1A2	"	"	"
			33										"									"	"	2A2	"	"	"
			34										"									"	"	3A2	"	"	"
			35										"									"	"	4A2	"	"	"
			36										"									"	"	$\overline{OE2}$	"	"	"
	I_{IH2}	3010	37	7.0 V									"									"	"	$\overline{OE1}$		100	"
			38		7.0 V		7.0 V		7.0 V		7.0 V		"	7.0 V		7.0 V		7.0 V		7.0 V		7.0 V	"	1A1	"	"	"
			39										"									"	"	2A1	"	"	"
			40										"									"	"	3A1	"	"	"
			41										"									"	"	4A1	"	"	"
			42										"									"	"	1A2	"	"	"
			43										"									"	"	2A2	"	"	"
			44										"									"	"	3A2	"	"	"
			45										"									"	"	4A2	"	"	"
			46										"									"	"	$\overline{OE2}$	"	"	"
	I_{IL1}	3009	47		0.5 V		0.5 V		0.5 V		0.5 V		"	0.5 V		0.5 V		0.5 V		0.5 V			"	1A1	1/	1/	mA
			48										"										"	2A1	"	"	"
			49										"										"	3A1	"	"	"
			50										"										"	4A1	"	"	"
			51										"										"	1A2	"	"	"
			52										"										"	2A2	"	"	"
			53										"										"	3A2	"	"	"
			54										"										"	4A2	"	"	"
	I_{IL2} I_{IL2}	3009 3009	55 56	0.5 V									"									0.5 V	"	$\overline{OE1}$ $\overline{OE2}$	"	"	"

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases R, S, 2, X Test no.	1 $\overline{OE1}$	2 1A1	3 4Y2	4 2A1	5 3Y2	6 3A1	7 2Y2	8 4A1	9 1Y2	10 GND	11 1A2	12 4Y1	13 2A2	14 3Y1	15 3A2	16 2Y1	17 4A2	18 1Y1	19 $\overline{OE2}$	20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	I_{OZH}		57	2.0 V	0.0 V		0.0 V		0.0 V		0.0 V		GND				2.4 V		2.4 V		2.4 V		5.5 V	1Y1		50	µA
			58	"									"										"	2Y1		"	"
			59	"									"										"	3Y1		"	"
			60	"									"										"	4Y1		"	"
			61										"									2.0 V	"	1Y2		"	"
			62							2.4 V			"					0.0 V				"	"	2Y2		"	"
			63			2.4 V		2.4 V				2.4 V	"	0.0 V		0.0 V						"	"	3Y2		"	"
			64										"		2.4 V					0.0 V		"	"	4Y2		"	"
	I_{OZL}		65	2.0 V	4.5 V		4.5 V		4.5 V		4.5 V		"				0.5 V		0.5 V		0.5 V		"	1Y1		-50	"
			66	"									"										"	2Y1		"	"
			67	"									"										"	3Y1		"	"
			68	"									"										"	4Y1		"	"
			69										"									2.0 V	"	1Y2		"	"
			70							0.5 V			"					4.5 V				"	"	2Y2		"	"
			71			0.5 V		0.5 V				0.5 V	"	4.5 V		4.5 V						"	"	3Y2		"	"
			72										"		0.5 V					4.5 V		"	"	4Y2		"	"
	I_{OS}	3011	73	0.0 V	4.5 V		4.5 V		4.5 V		4.5 V		"				0.0 V		0.0 V		0.0 V		"	1Y1	-100	-225 2/	mA
			74	"									"										"	2Y1	"	"	"
			75	"									"										"	3Y1	"	"	"
			76	"									"										"	4Y1	"	"	"
			77										"									0.0 V	"	1Y2	"	"	"
			78							0.0 V			"					4.5 V				"	"	2Y2	"	"	"
			79			0.0 V		0.0 V				0.0 V	"	4.5 V		4.5 V						"	"	3Y2	"	"	"
			80										"		0.0 V					4.5 V		"	"	4Y2	"	"	"
	I_{OD}		81	GND	GND		GND		GND		GND		"				2.5 V		2.5 V		2.5 V		4.5 V	1Y1	65		"
			82	"									"										"	2Y1	"		"
			83	"									"										"	3Y1	"		"
			84	"									"										"	4Y1	"		"
			85										"									GND	"	1Y2	"		"
			86							2.5 V			"					GND				"	"	2Y2	"		"
			87			2.5 V		2.5 V				2.5 V	"	GND		GND						"	"	3Y2	"		"
			88										"		2.5 V					GND		"	"	4Y2	"		"
	I_{CCH}	3005	89	0.0 V	4.5 V		4.5 V		4.5 V		4.5 V		"	4.5 V		4.5 V		4.5 V		4.5 V		0.0 V	5.5 V	V_{CC}		60	"
	I_{CCL}	3005	90	0.0 V	0.0 V		0.0 V		0.0 V		0.0 V		"	0.0 V		0.0 V		0.0 V		0.0 V		0.0 V	"	V_{CC}		90	"
	I_{CCZ}	3005	91	4.5 V									"									4.5 V	"	V_{CC}		90	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																										
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																										
7 3/ T_C = +25°C	Functional tests	3014 3014	92	B	B	L	B	L	B	L	B	L	GND	B	L	B	L	B	L	B	L	B	4/ $\overline{4}$/	All outputs			
			93	B	A	H	A	H	A	H	A	H	"	A	H	A	H	A	H	A	H	B	All outputs				
9 T_C = +25°C	t_{PLH2}	3003 (Fig. 4)	94	0.0 V	IN		IN		IN		IN		"				OUT		OUT		OUT		5.0 V	1A1 to 1Y1	1.5	5.2	ns
			95	"									"										"	2A1 to 2Y1	"	"	"
			96	"									"										"	3A1 to 3Y1	"	"	"
			97										"									0.0 V	"	4A1 to 4Y1	"	"	"
			98							OUT			"					IN				"	"	1A2 to 1Y2	"	"	"
			99										"									"	"	2A2 to 2Y2	"	"	"
			100										"									"	"	3A2 to 3Y2	"	"	"
			101			OUT		OUT				OUT	"	IN		IN				IN			"	4A2 to 4Y2	"	"	"

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases R, S, 2, X Test no.	1 $\overline{OE1}$	2 1A1	3 4Y2	4 2A1	5 3Y2	6 3A1	7 2Y2	8 4A1	9 1Y2	10 GND	11 1A2	12 4Y1	13 2A2	14 3Y1	15 3A2	16 2Y1	17 4A2	18 1Y1	19 $\overline{OE2}$	20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
9 T_C = +25°C	t_{PHL2}	3003 (Fig. 4)	102	0.0 V	IN	OUT	IN	OUT	IN	OUT	IN	OUT	GND	IN	OUT	IN	OUT	IN	OUT	IN	OUT	0.0 V	5.0 V	1A1 to 1Y1	1.5	6.0	ns
			103	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	2A1 to 2Y1	"	"	"
			104	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	3A1 to 3Y1	"	"	"
			105	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	4A1 to 4Y1	"	"	"
			106	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	1A2 to 1Y2	"	"	"
			107	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	2A2 to 2Y2	"	"	"
			108	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	3A2 to 3Y2	"	"	"
			109	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	4A2 to 4Y2	"	"	"
	t_{PHZ2}	3003 (Fig. 4)	110	IN	2.7 V		2.7 V		2.7 V		2.7 V		"	2.7 V		2.7 V		2.7 V		2.7 V		IN	"	$\overline{OE1}$ to 1Y1	2.0	"	"
			111	"	"		"		"		"		"	"		"		"		"		"	"	$\overline{OE1}$ to 2Y1	"	"	"
			112	"	"		"		"		"		"	"		"		"		"		"	"	$\overline{OE1}$ to 3Y1	"	"	"
			113	"	"		"		"		"		"	"		"		"		"		"	"	$\overline{OE1}$ to 4Y1	"	"	"
			114	"	"		"		"		"		"	"		"		"		"		"	"	$\overline{OE2}$ to 1Y2	"	"	"
			115	"	"		"		"		"		"	"		"		"		"		"	"	$\overline{OE2}$ to 2Y2	"	"	"
			116	"	"		"		"		"		"	"		"		"		"		"	"	$\overline{OE2}$ to 3Y2	"	"	"
			117	"	"		"		"		"		"	"		"		"		"		"	"	$\overline{OE2}$ to 4Y2	"	"	"
	t_{PLZ2}	3003 (Fig. 4)	118	IN	0.0 V		0.0 V		0.0 V		0.0 V		"	0.0 V		0.0 V		0.0 V		0.0 V		IN	"	$\overline{OE1}$ to 1Y1	"	"	"
			119	"	"		"		"		"		"	"		"		"		"		"	"	$\overline{OE1}$ to 2Y1	"	"	"
			120	"	"		"		"		"		"	"		"		"		"		"	"	$\overline{OE1}$ to 3Y1	"	"	"
			121	"	"		"		"		"		"	"		"		"		"		"	"	$\overline{OE1}$ to 4Y1	"	"	"
			122	"	"		"		"		"		"	"		"		"		"		"	"	$\overline{OE2}$ to 1Y2	"	"	"
			123	"	"		"		"		"		"	"		"		"		"		"	"	$\overline{OE2}$ to 2Y2	"	"	"
			124	"	"		"		"		"		"	"		"		"		"		"	"	$\overline{OE2}$ to 3Y2	"	"	"
			125	"	"		"		"		"		"	"		"		"		"		"	"	$\overline{OE2}$ to 4Y2	"	"	"
	t_{PZH2}	3003 (Fig. 4)	126	IN	2.7 V		2.7 V		2.7 V		2.7 V		"	2.7 V		2.7 V		2.7 V		2.7 V		IN	"	$\overline{OE1}$ to 1Y1	"	5.7	"
			127	"	"		"		"		"		"	"		"		"		"		"	"	$\overline{OE1}$ to 2Y1	"	"	"
			128	"	"		"		"		"		"	"		"		"		"		"	"	$\overline{OE1}$ to 3Y1	"	"	"
			129	"	"		"		"		"		"	"		"		"		"		"	"	$\overline{OE1}$ to 4Y1	"	"	"
			130	"	"		"		"		"		"	"		"		"		"		"	"	$\overline{OE2}$ to 1Y2	"	"	"
			131	"	"		"		"		"		"	"		"		"		"		"	"	$\overline{OE2}$ to 2Y2	"	"	"
			132	"	"		"		"		"		"	"		"		"		"		"	"	$\overline{OE2}$ to 3Y2	"	"	"
			133	"	"		"		"		"		"	"		"		"		"		"	"	$\overline{OE2}$ to 4Y2	"	"	"
	t_{PZL2}	3003 (Fig. 4)	134	IN	0.0 V		0.0 V		0.0 V		0.0 V		"	0.0 V		0.0 V		0.0 V		0.0 V		IN	"	$\overline{OE1}$ to 1Y1	"	7.0	"
			135	"	"		"		"		"		"	"		"		"		"		"	"	$\overline{OE1}$ to 2Y1	"	"	"
			136	"	"		"		"		"		"	"		"		"		"		"	"	$\overline{OE1}$ to 3Y1	"	"	"
			137	"	"		"		"		"		"	"		"		"		"		"	"	$\overline{OE1}$ to 4Y1	"	"	"
			138	"	"		"		"		"		"	"		"		"		"		"	"	$\overline{OE2}$ to 1Y2	"	"	"
			139	"	"		"		"		"		"	"		"		"		"		"	"	$\overline{OE2}$ to 2Y2	"	"	"
			140	"	"		"		"		"		"	"		"		"		"		"	"	$\overline{OE2}$ to 3Y2	"	"	"
			141	"	"		"		"		"		"	"		"		"		"		"	"	$\overline{OE2}$ to 4Y2	"	"	"
10 T_C = +125°C	t_{PLH2}	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C.																							1.0	6.5	"
	t_{PHL2}																								1.0	7.0	"
	t_{PHZ2}																								2.0	7.0	"
	t_{PLZ2}																								"	7.5	"
	t_{PZH2}																								"	7.0	"
	t_{PZL2}																								"	8.5	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																										

1/ I_{IL} limits shall be as follows:

Test	Min/Max limits in (mA) Circuits A	B	C
I_{IL1}	-.67/-1.6	-.03/-1.6	-.03/-1.6
I_{IL2}	-.42/-1.0	-.03/-1.0	-.06/-1.0

2/ I_{OS} limits (mA) min/max values for CKT B, -100/-325.

3/ A = 2.4 V minimum; B = 0.55 V or GND, H ≥1.5 V, L ≤ 1.5 V.

4/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

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TABLE III. Group A inspection for device type 04.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases R, S, 2, X	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Limits		Unit
			Test no.	$\overline{OE1}$	1A	2A	3A	4A	5A	6A	7A	8A	GND	8$\overline{Y}$	7$\overline{Y}$	6$\overline{Y}$	5$\overline{Y}$	4$\overline{Y}$	3$\overline{Y}$	2$\overline{Y}$	1$\overline{Y}$	$\overline{OE2}$	V_{CC}		Min	Max	
1 T_C = +25°C	V_{OH}	3006	1	0.8 V									GND	-3.0 mA								0.8 V	4.5 V	8$\overline{Y}$	2.4		V
			2	"									"		-3.0 mA							"	"	7$\overline{Y}$	"		"
			3	"									"			-3.0 mA						"	"	6$\overline{Y}$	"		"
			4	"									"				-3.0 mA					"	"	5$\overline{Y}$	"		"
			5	"									"					-3.0 mA				"	"	4$\overline{Y}$	"		"
			6	"									"						-3.0 mA			"	"	3$\overline{Y}$	"		"
			7	"									"							-3.0 mA		"	"	2$\overline{Y}$	"		"
			8	"	0.8 V								"								-3.0 mA	"	"	1$\overline{Y}$	"		"
	V_{OL}	3007	9	"									"	48 mA								"	"	8$\overline{Y}$		0.55	"
			10	"									"		48 mA							"	"	7$\overline{Y}$		"	"
			11	"									"			48 mA						"	"	6$\overline{Y}$		"	"
			12	"									"				48 mA					"	"	5$\overline{Y}$		"	"
			13	"									"					48 mA				"	"	4$\overline{Y}$		"	"
			14	"									"						48 mA			"	"	3$\overline{Y}$		"	"
			15	"									"							48 mA		"	"	2$\overline{Y}$		"	"
			16	"	2.0 V								"								48 mA	"	"	1$\overline{Y}$		"	"
	V_{IC}		17	-18 mA									"									0.0 V	"	$\overline{OE1}$		-1.2	"
			18	5.5 V	-18 mA								"									5.5 V	"	1A		"	"
			19	"		-18 mA							"									"	"	2A		"	"
			20	"			-18 mA						"									"	"	3A		"	"
			21	"				-18 mA					"									"	"	4A		"	"
			22	"					-18 mA				"									"	"	5A		"	"
			23	"						-18 mA			"									"	"	6A		"	"
			24	"							-18 mA		"									"	"	7A		"	"
			25	"								-18 mA	"									"	"	8A		"	"
			26	0.0 V									"									-18 mA	"	$\overline{OE2}$		"	"
	I_{IH1}	3010	27	2.7 V									"									0.0 V	5.5 V	$\overline{OE1}$		20	µA
			28	5.5 V	2.7 V								"									5.5 V	"	1A		"	"
			29	"		2.7 V							"									"	"	2A		"	"
			30	"			2.7 V						"									"	"	3A		"	"
			31	"				2.7 V					"									"	"	4A		"	"
			32	"					2.7 V				"									"	"	5A		"	"
			33	"						2.7 V			"									"	"	6A		"	"
			34	"							2.7 V		"									"	"	7A		"	"
			35	"								2.7 V	"									"	"	8A		"	"
			36	0.0 V									"									2.7 V	"	$\overline{OE2}$		"	"
	I_{IH2}	3010	37	7.0 V									"									0.0 V	"	$\overline{OE1}$		100	µA
			38	5.5 V	7.0 V								"									5.5 V	"	1A		"	"
			39	"		7.0 V							"									"	"	2A		"	"
			40	"			7.0 V						"									"	"	3A		"	"
			41	"				7.0 V					"									"	"	4A		"	"
			42	"					7.0 V				"									"	"	5A		"	"
			43	"						7.0 V			"									"	"	6A		"	"
			44	"							7.0 V		"									"	"	7A		"	"
			45	"								7.0 V	"									"	"	8A		"	"
			46	0.0 V									"									2.7 V	"	$\overline{OE2}$		"	"
	I_{IL}	3009	47	0.5 V									"									5.5 V	"	$\overline{OE1}$	1/	1/	mA
			48	0.0 V	0.5 V								"									0.0 V	"	1A	"	"	"
			49	"		0.5 V							"									"	"	2A	"	"	"
			50	"			0.5 V						"									"	"	3A	"	"	"
			51	"				0.5 V					"									"	"	4A	"	"	"
			52	"					0.5 V				"									"	"	5A	"	"	"
			53	"						0.5 V			"									"	"	6A	"	"	"
			54	"							0.5 V		"									"	"	7A	"	"	"
			55	"								0.5 V	"									"	"	8A	"	"	"
			56	5.5 V									"									0.5 V	"	$\overline{OE2}$	"	"	"

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases R, S, 2, X Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Limits Min	Limits Max	Unit
				$\overline{OE1}$	1A	2A	3A	4A	5A	6A	7A	8A	GND	8$\overline{Y}$	7$\overline{Y}$	6$\overline{Y}$	5$\overline{Y}$	4$\overline{Y}$	3$\overline{Y}$	2$\overline{Y}$	1$\overline{Y}$	$\overline{OE2}$	V_{CC}				
1 T_C = +25°C	I_{OS}	3011	57	0.0 V										0.0 V										8$\overline{Y}$	-100	-225	mA
			58	"											0.0 V									7$\overline{Y}$	"	"	"
			59	"												0.0 V								6$\overline{Y}$	"	"	"
			60	"				0.0 V									0.0 V								5$\overline{Y}$	"	"	"
			61	"			0.0 V		0.0 V									0.0 V							4$\overline{Y}$	"	"	"
			62	"		0.0 V				0.0 V									0.0 V						3$\overline{Y}$	"	"	"
			63	"	0.0 V						0.0 V									0.0 V						2$\overline{Y}$	"	"	"
			64	"								0.0 V									0.0 V					1$\overline{Y}$	"	"	"
	I_{CCH} I_{CCL} I_{CCZ}	3005	65	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	"										"	"	V_{CC}		20	"
			66	0.0 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"										"	"	V_{CC}		75	"
			67	5.5 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	"										"	"	V_{CC}		45	"
	I_{OZH}		68	2.0 V										2.7 V										8$\overline{Y}$		50	µA
			69	"											2.7 V									7$\overline{Y}$		"	"
			70	"												2.7 V								6$\overline{Y}$		"	"
			71	"													2.7 V							5$\overline{Y}$		"	"
			72	"														2.7 V						4$\overline{Y}$		"	"
			73	"															2.7 V					3$\overline{Y}$		"	"
			74	"																2.7 V				2$\overline{Y}$		"	"
			75	"																	2.7 V			1$\overline{Y}$		"	"
			76	0.0 V	5.5 V									2.7 V								2.0 V	"	8$\overline{Y}$		"	"
			77	"		5.5 V									2.7 V							"	"	7$\overline{Y}$		"	"
			78	"			5.5 V									2.7 V						"	"	6$\overline{Y}$		"	"
			79	"				5.5 V		5.5 V							2.7 V					"	"	5$\overline{Y}$		"	"
			80	"					5.5 V		5.5 V							2.7 V				"	"	4$\overline{Y}$		"	"
			81	"			5.5 V					5.5 V							2.7 V			"	"	3$\overline{Y}$		"	"
			82	"		5.5 V														2.7 V		"	"	2$\overline{Y}$		"	"
			83	"	5.5 V																2.7 V	"	"	1$\overline{Y}$		"	"
	I_{OZL}		84	2.0 V										0.5 V										8$\overline{Y}$		-50	µA
			85	"											0.5 V									7$\overline{Y}$		"	"
			86	"												0.5 V								6$\overline{Y}$		"	"
			87	"													0.5 V							5$\overline{Y}$		"	"
			88	"														0.5 V						4$\overline{Y}$		"	"
			89	"															0.5 V					3$\overline{Y}$		"	"
			90	"																0.5 V				2$\overline{Y}$		"	"
			91	"																	0.5 V			1$\overline{Y}$		"	"
			92	0.0 V	0.0 V									0.5 V								0.0 V	"	8$\overline{Y}$		"	"
			93	"		0.0 V									0.5 V							"	"	7$\overline{Y}$		"	"
			94	"			0.0 V									0.5 V						"	"	6$\overline{Y}$		"	"
			95	"				0.0 V		0.0 V							0.5 V					"	"	5$\overline{Y}$		"	"
			96	"					0.0 V		0.0 V							0.5 V				"	"	4$\overline{Y}$		"	"
			97	"			0.0 V					0.0 V							0.5 V			"	"	3$\overline{Y}$		"	"
			98	"		0.0 V														0.5 V		"	"	2$\overline{Y}$		"	"
			99	"	0.0 V																0.5 V	"	"	1$\overline{Y}$		"	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																										
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																										
7 T_C = +25°C	Functional tests 3/	3014	100	B	B	B	B	B	B	B	B	B	GND	H	H	H	H	H	H	H	H	B	2/	All outputs			
			101	A	A	A	A	A	A	A	A	A	"	L	L	L	L	L	L	L	L	"	"				
			102	B	A	B	B	A	B	A	B	B	"	L	H	L	H	L	L	L	L	"	"				
			103	B	B	A	A	B	A	B	A	B	"	H	L	L	L	H	L	L	H	"	"				
9 T_C = +25°C	t_{PLH3}	3003	104	0.0V	IN	IN	IN	IN	IN	IN	IN	IN	"	OUT								0.0 V	5.0 V	1A to 1$\overline{Y}$	1.5	5.0	ns
			105	"									"		OUT							"	"	2A to 2$\overline{Y}$	"	"	"
			106	"									"			OUT						"	"	3A to 3$\overline{Y}$	"	"	"
			107	"									"				OUT					"	"	4A to 4$\overline{Y}$	"	"	"
			108	"									"					OUT				"	"	5A to 5$\overline{Y}$	"	"	"
			109	"									"						OUT			"	"	6A to 6$\overline{Y}$	"	"	"
			110	"									"							OUT		"	"	7A to 7$\overline{Y}$	"	"	"
			111	"									"								OUT	"	"	8A to 8$\overline{Y}$	"	"	"

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases R, S, 2, X Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Limits Min	Limits Max	Unit
				$\overline{OE1}$	1A	2A	3A	4A	5A	6A	7A	8A	GND	$8\overline{Y}$	$7\overline{Y}$	$6\overline{Y}$	$5\overline{Y}$	$4\overline{Y}$	$3\overline{Y}$	$2\overline{Y}$	$1\overline{Y}$	$\overline{OE2}$	V_{CC}				
9 T_C = +25°C	t_{PHL3}	3003	112 113 114 115 116 117 118 119	0.0 V " " " " " " "	IN	IN	IN	IN	IN	IN	IN	IN	GND " " " " " " "	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	0.0 V " " " " " " "	5.0 V " " " " " " "	1A to $1\overline{Y}$ 2A to $2\overline{Y}$ 3A to $3\overline{Y}$ 4A to $4\overline{Y}$ 5A to $5\overline{Y}$ 6A to $6\overline{Y}$ 7A to $7\overline{Y}$ 8A to $8\overline{Y}$	1.0 " " " " " " "	4.0 " " " " " " "	ns " " " " " " "
	t_{PZH4}		120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135	IN " " " " " " " 0.0 V " " " " " " "	0.0 V 0.0 V	0.0 V 0.0 V	0.0 V 0.0 V	0.0 V 0.0 V	0.0 V 0.0 V	0.0 V 0.0 V	0.0 V 0.0 V	0.0 V 0.0 V	0.0 V 0.0 V	" " " " " " " " " " " " " " " "	OUT OUT	OUT OUT	OUT OUT	OUT OUT	OUT OUT	OUT OUT	OUT OUT	OUT OUT	" " " " " " " " IN " " " " " " "	" " " " " " " " " " " " " " " "	$\overline{OE1}$ to $1\overline{Y}$ $\overline{OE1}$ to $2\overline{Y}$ $\overline{OE1}$ to $3\overline{Y}$ $\overline{OE1}$ to $4\overline{Y}$ $\overline{OE1}$ to $5\overline{Y}$ $\overline{OE1}$ to $6\overline{Y}$ $\overline{OE1}$ to $7\overline{Y}$ $\overline{OE1}$ to $8\overline{Y}$ $\overline{OE2}$ to $1\overline{Y}$ $\overline{OE2}$ to $2\overline{Y}$ $\overline{OE2}$ to $3\overline{Y}$ $\overline{OE2}$ to $4\overline{Y}$ $\overline{OE2}$ to $5\overline{Y}$ $\overline{OE2}$ to $6\overline{Y}$ $\overline{OE2}$ to $7\overline{Y}$ $\overline{OE2}$ to $8\overline{Y}$	2.5 " " " " " " " " " " " " " " "	8.0 " " " " " " " " " " " " " " "	" " " " " " " " " " " " " " " "
	t_{PZL4}	3003	136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151	IN " " " " " " " 0.0 V " " " " " " "	2.7 V 2.7 V	2.7 V 2.7 V	2.7 V 2.7 V	2.7 V 2.7 V	2.7 V 2.7 V	2.7 V 2.7 V	2.7 V 2.7 V	2.7 V 2.7 V	" " " " " " " " " " " " " " " "	OUT OUT	OUT OUT	OUT OUT	OUT OUT	OUT OUT	OUT OUT	OUT OUT	OUT OUT	" " " " " " " " IN " " " " " " "	" " " " " " " " " " " " " " " "	$\overline{OE1}$ to $1\overline{Y}$ $\overline{OE1}$ to $2\overline{Y}$ $\overline{OE1}$ to $3\overline{Y}$ $\overline{OE1}$ to $4\overline{Y}$ $\overline{OE1}$ to $5\overline{Y}$ $\overline{OE1}$ to $6\overline{Y}$ $\overline{OE1}$ to $7\overline{Y}$ $\overline{OE1}$ to $8\overline{Y}$ $\overline{OE2}$ to $1\overline{Y}$ $\overline{OE2}$ to $2\overline{Y}$ $\overline{OE2}$ to $3\overline{Y}$ $\overline{OE2}$ to $4\overline{Y}$ $\overline{OE2}$ to $5\overline{Y}$ $\overline{OE2}$ to $6\overline{Y}$ $\overline{OE2}$ to $7\overline{Y}$ $\overline{OE2}$ to $8\overline{Y}$	3.5 " " " " " " " " " " " " " " "	10.0 " " " " " " " " " " " " " " "	" " " " " " " " " " " " " " " "
	t_{PHZ4}	3003	152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167	IN " " " " " " " 0.0 V " " " " " " "	0.0 V 0.0 V	0.0 V 0.0 V	0.0 V 0.0 V	0.0 V 0.0 V	0.0 V 0.0 V	0.0 V 0.0 V	0.0 V 0.0 V	0.0 V 0.0 V	" " " " " " " " " " " " " " " "	OUT OUT	OUT OUT	OUT OUT	OUT OUT	OUT OUT	OUT OUT	OUT OUT	OUT OUT	" " " " " " " " IN " " " " " " "	" " " " " " " " " " " " " " " "	$\overline{OE1}$ to $1\overline{Y}$ $\overline{OE1}$ to $2\overline{Y}$ $\overline{OE1}$ to $3\overline{Y}$ $\overline{OE1}$ to $4\overline{Y}$ $\overline{OE1}$ to $5\overline{Y}$ $\overline{OE1}$ to $6\overline{Y}$ $\overline{OE1}$ to $7\overline{Y}$ $\overline{OE1}$ to $8\overline{Y}$ $\overline{OE2}$ to $1\overline{Y}$ $\overline{OE2}$ to $2\overline{Y}$ $\overline{OE2}$ to $3\overline{Y}$ $\overline{OE2}$ to $4\overline{Y}$ $\overline{OE2}$ to $5\overline{Y}$ $\overline{OE2}$ to $6\overline{Y}$ $\overline{OE2}$ to $7\overline{Y}$ $\overline{OE2}$ to $8\overline{Y}$	1.5 " " " " " " " " " " " " " " "	6.0 " " " " " " " " " " " " " " "	" " " " " " " " " " " " " " " "

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≤2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases R, S, 2, X Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Limits Min	Limits Max	Unit
				$\overline{OE1}$	1A	2A	3A	4A	5A	6A	7A	8A	GND	8$\overline{Y}$	7$\overline{Y}$	6$\overline{Y}$	5$\overline{Y}$	4$\overline{Y}$	3$\overline{Y}$	2$\overline{Y}$	1$\overline{Y}$	$\overline{OE2}$	V_{CC}				
9 T_C = +25°C	t_{PLZ4}	3003	168	IN	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	GND									0.0 V	5.0 V	$\overline{OE1}$ to 1$\overline{Y}$	1.0	5.5	ns
			169	"									"									"	"	$\overline{OE1}$ to 2$\overline{Y}$	"	"	"
			170	"									"									"	"	$\overline{OE1}$ to 3$\overline{Y}$	"	"	"
			171	"									"				OUT					"	"	$\overline{OE1}$ to 4$\overline{Y}$	"	"	"
			172	"									"			OUT						"	"	$\overline{OE1}$ to 5$\overline{Y}$	"	"	"
			173	"									"		OUT							"	"	$\overline{OE1}$ to 6$\overline{Y}$	"	"	"
			174	"									"	OUT								"	"	$\overline{OE1}$ to 7$\overline{Y}$	"	"	"
			175	"									"									"	"	$\overline{OE1}$ to 8$\overline{Y}$	"	"	"
			176	0.0 V	2.7 V								"									IN	"	$\overline{OE2}$ to 1$\overline{Y}$	"	"	"
			177	"		2.7 V							"								OUT	"	"	$\overline{OE2}$ to 2$\overline{Y}$	"	"	"
			178	"			2.7 V						"							OUT		"	"	$\overline{OE2}$ to 3$\overline{Y}$	"	"	"
			179	"				2.7 V					"						OUT			"	"	$\overline{OE2}$ to 4$\overline{Y}$	"	"	"
			180	"					2.7 V				"					OUT				"	"	$\overline{OE2}$ to 5$\overline{Y}$	"	"	"
			181	"						2.7 V			"				OUT					"	"	$\overline{OE2}$ to 6$\overline{Y}$	"	"	"
			182	"							2.7 V		"			OUT						"	"	$\overline{OE2}$ to 7$\overline{Y}$	"	"	"
			183	"								2.7 V	"	OUT	OUT							"	"	$\overline{OE2}$ to 8$\overline{Y}$	"	"	"
10	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and use limits from table I.																										
11	Same tests and terminal conditions as for subgroup 10, except T_C = -55°C.																										

1/ I_{IL} limits shall be as follows:

Test	Min/Max limits in (mA) Circuits A	B	C	D
I_{IL}	-.25/-.60			

2/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

3/ A = 2.4 V minimum; B = 0.55 V or GND, H ≥ 1.5 V, L ≤ 1.5 V.

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4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for use for Government microcircuit applications (original equipment) and logistic purposes.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable, these requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND - - - - - - - - - - - - - Ground (zero voltage potential)
I_{IN} - - - - - - - - - - - - - Current flowing into an input terminal
V_{IN} - - - - - - - - - - - - - Voltage level at an input terminal

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) and lead finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

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6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-35810 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-35810 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54F240
02	54F241
03	54F244
04	54F540

6.6 Manufacturers' designation. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designation.

Device type	Manufacturer			
	Circuit A	Circuit B	Circuit C	Circuit D
	Fairchild Semiconductor	Motorola Incorporated	Signetics Corporation	Texas Instruments
01	X	X		
02	X	X		
03	X	X		
04	X			

6.7 Changes from previous issue. Asterisks are not used in this revision to identify change with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SM, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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